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Seeing is Not Believing: Inside the World of Deepfakes

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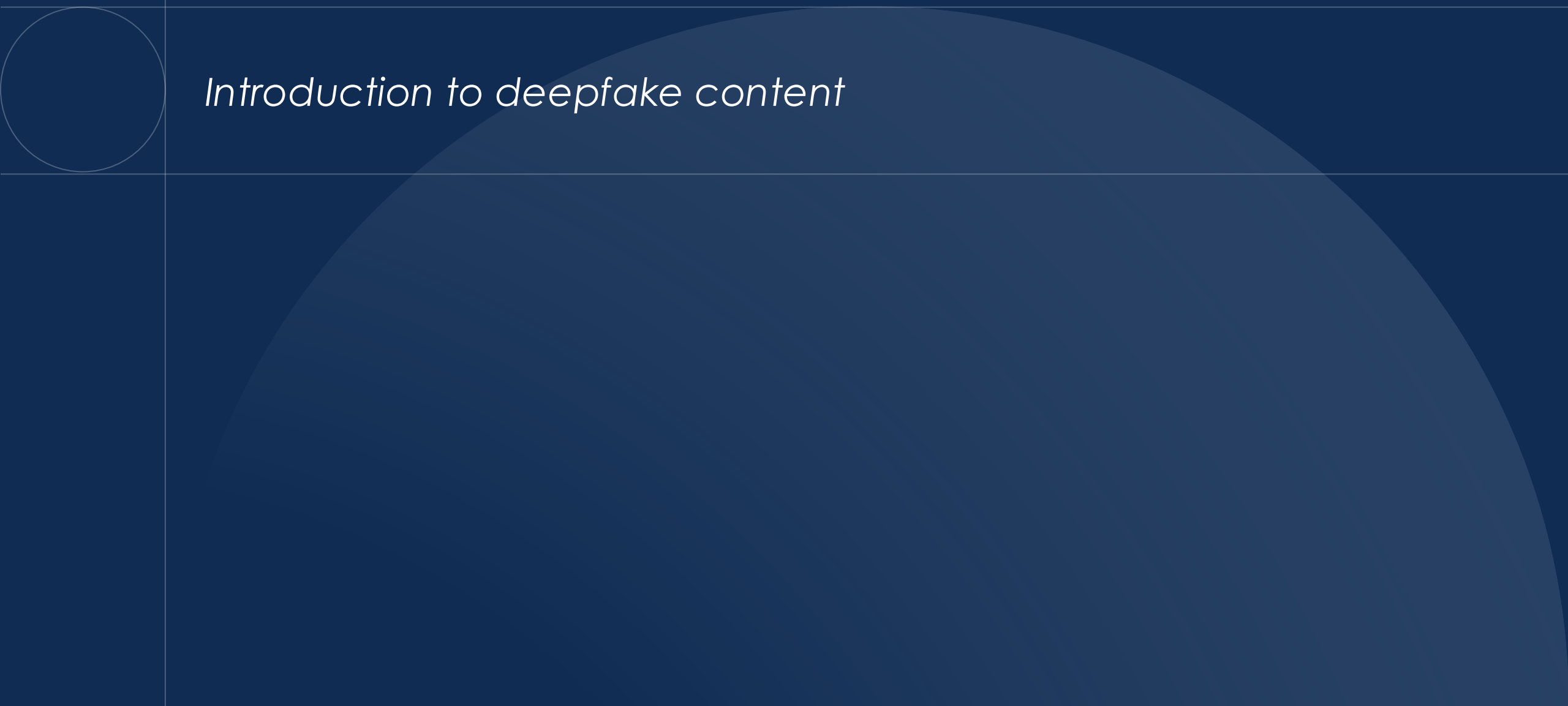
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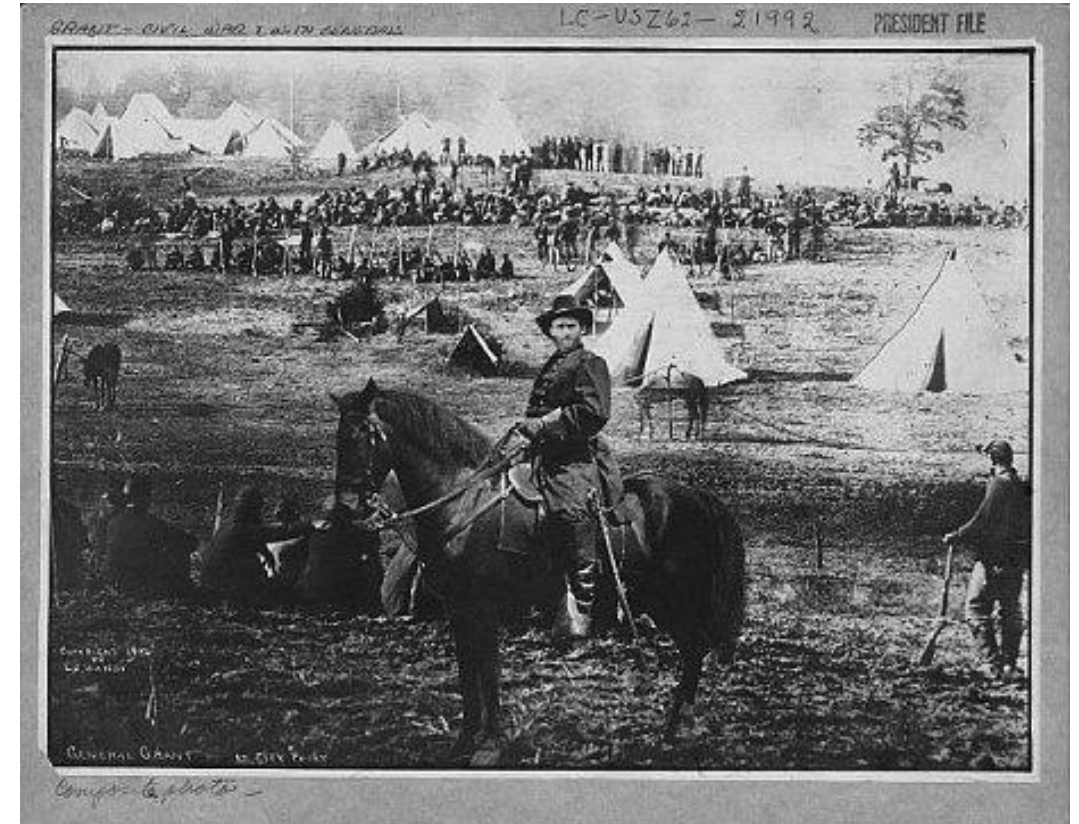
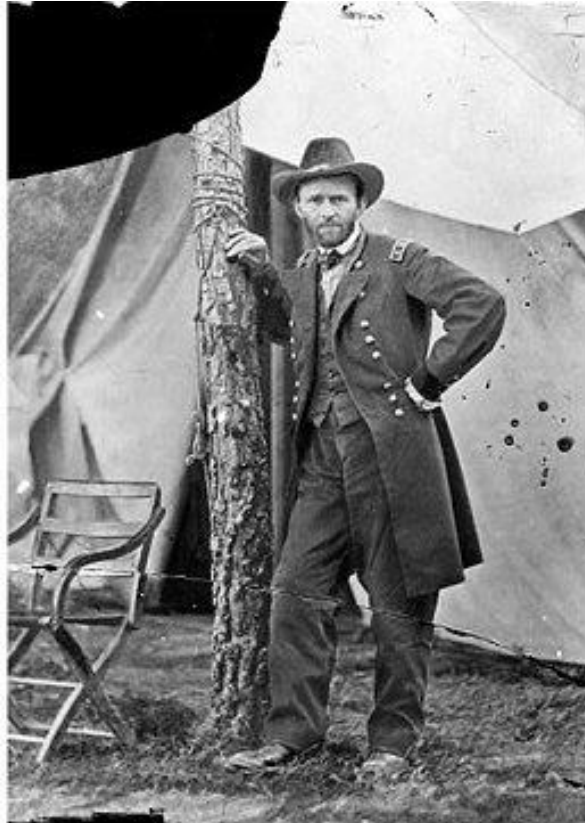


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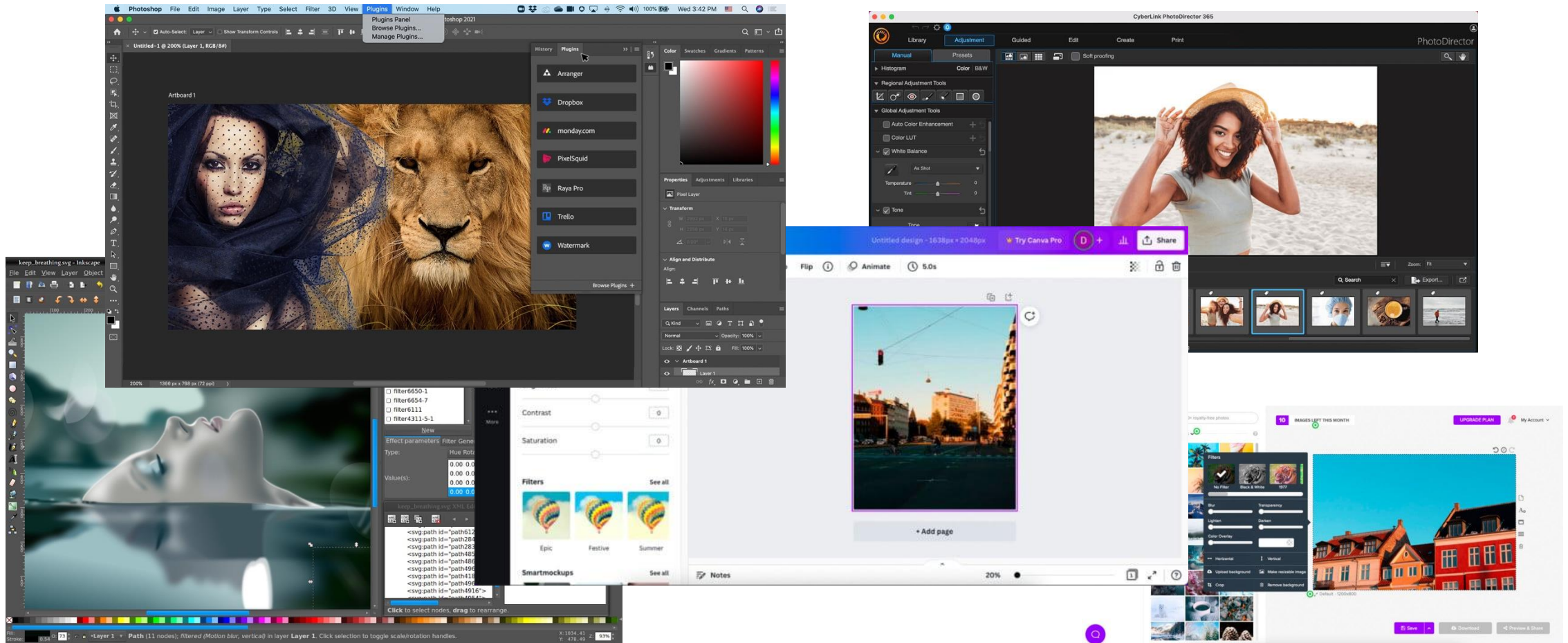
Introduction to deepfake content

Faking photos is nothing new...



General Ulysses Grant during the America Civil War (1861-1865)

... But today it is easier!



... But today it is easier!

2014



2015



GANs

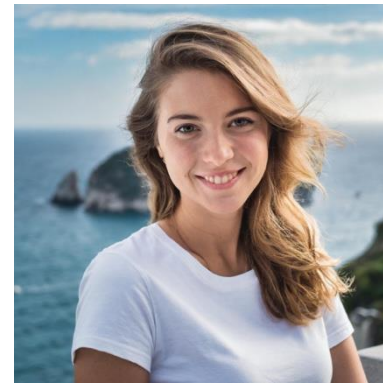
2017



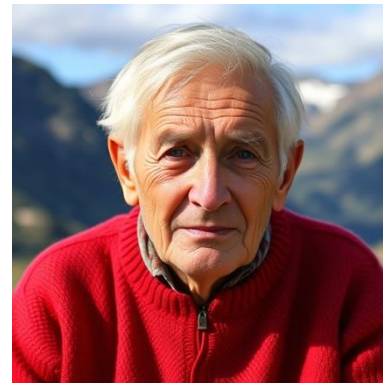
2020



2023



A blond girl with white t-shirt, sea in the background



An old man with red jumper, mountains in the background

2024



DIFFUSION MODELS

... But today it is easier!

Face-swap



... But today it is easier!

Face-swap



... But today it is easier!

Face-swap



... But today it is easier!

Face-swap







The other side of the coin



We cannot trust what we see

The spreading of fake content brings serious disinformation problems





Detecting traces of deepfakes



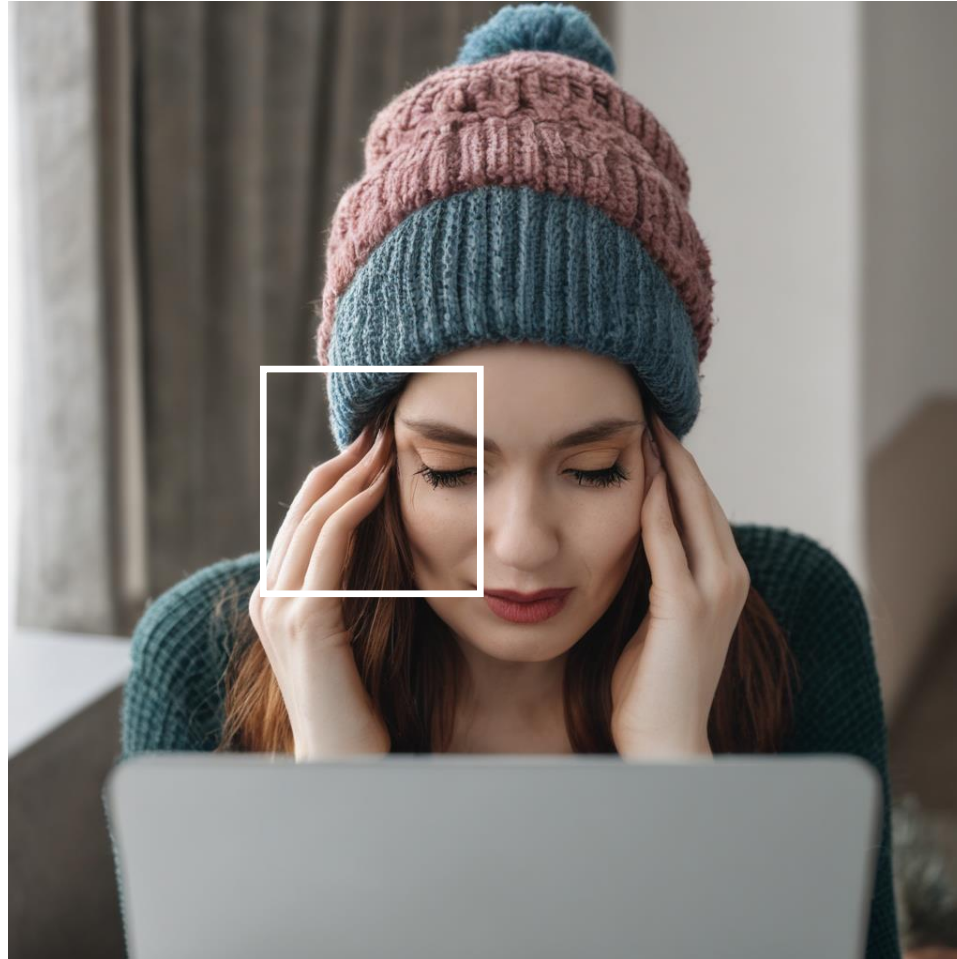
Detecting traces of deepfakes

High level traces

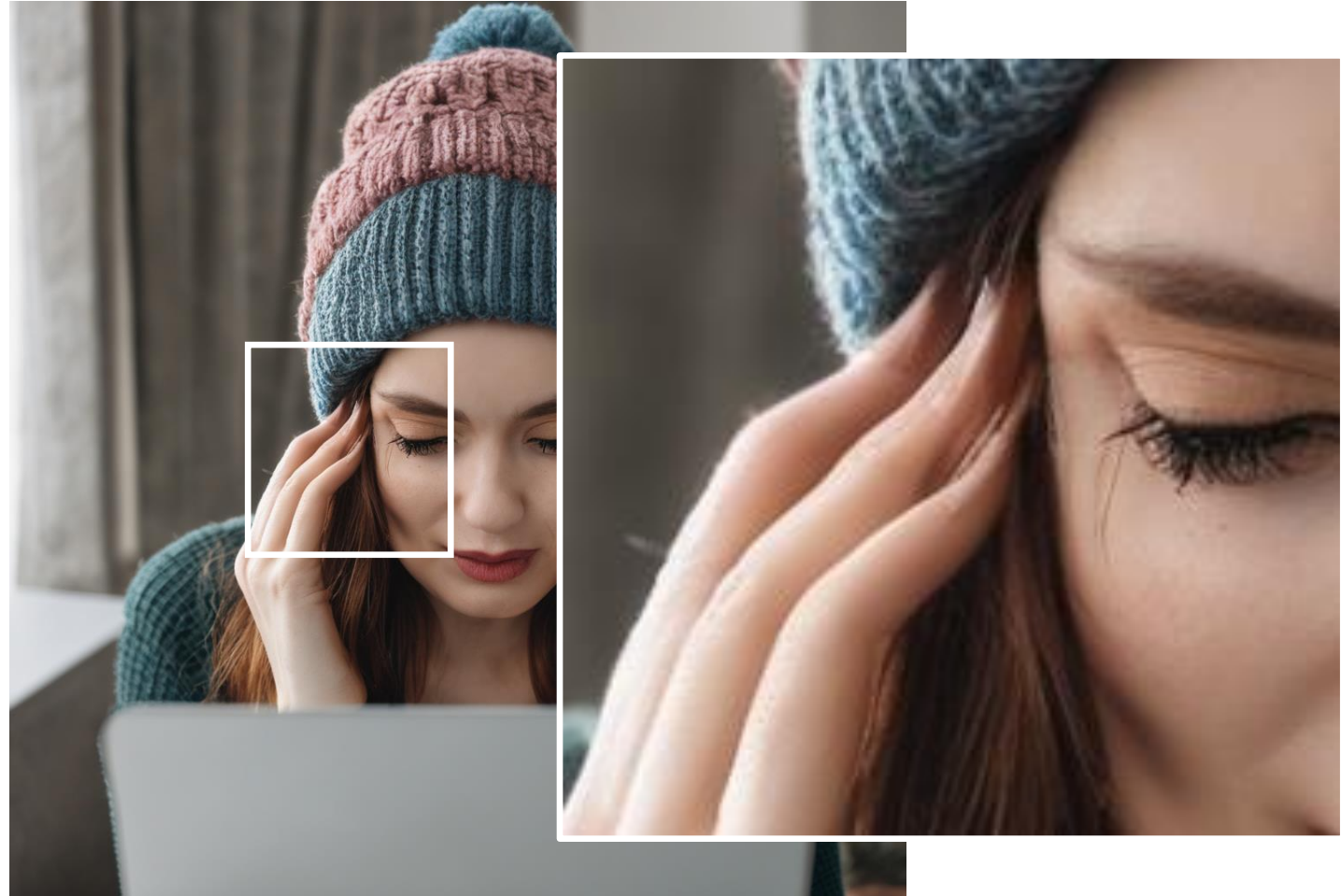
High level traces: semantic inconsistencies



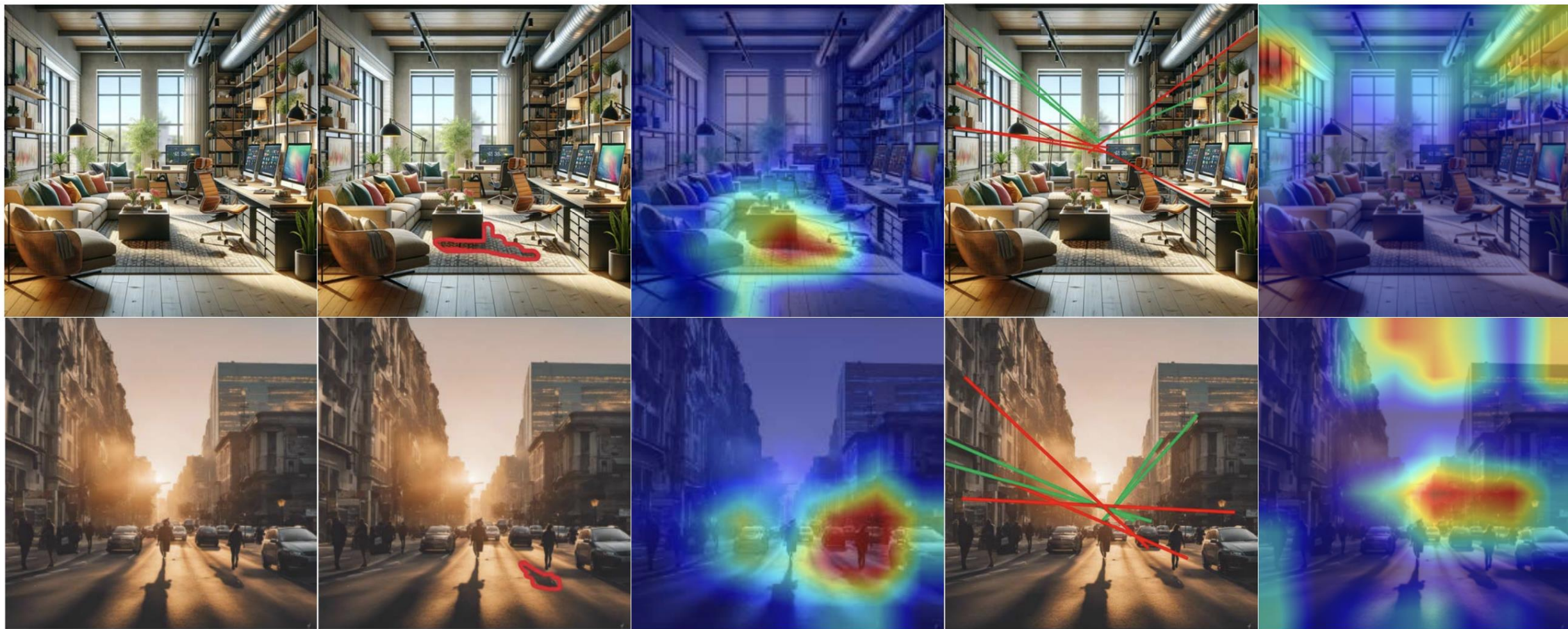
High level traces: semantic inconsistencies



High level traces: semantic inconsistencies



High level traces: shadow/perspective errors



Generated Image

Shadow Errors

Detected Shadow Errors

Vanishing Point Errors

Detected Perspective Errors

High level traces: temporal inconsistencies



High level traces: potential issues

Models are getting better → artifacts are going to be less visible



High level traces: potential issues

Models are getting better → artifacts are going to be less visible

Detectors can be biased by the training data content





Detecting traces of deepfakes

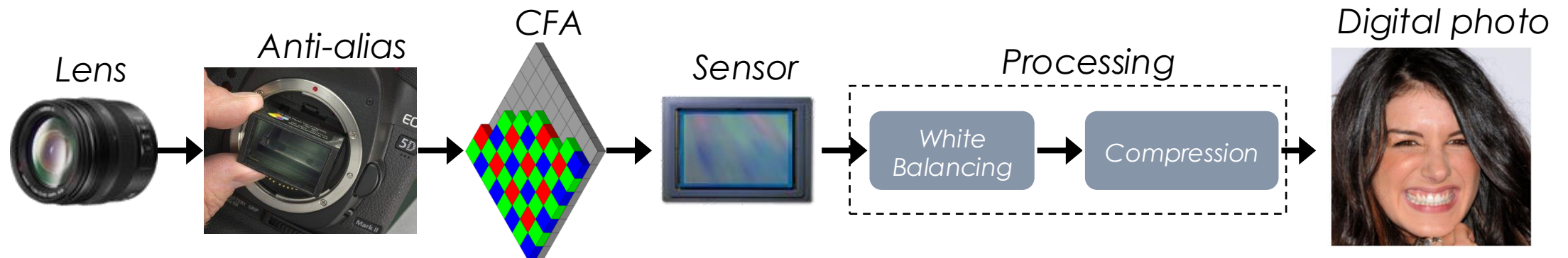
Low level traces

Low level traces:

Real images:

Specific traces left during the **image acquisition** process

- PRNU
- CFA
- White balancing
- Compression
- other processing operations...



Low level traces:

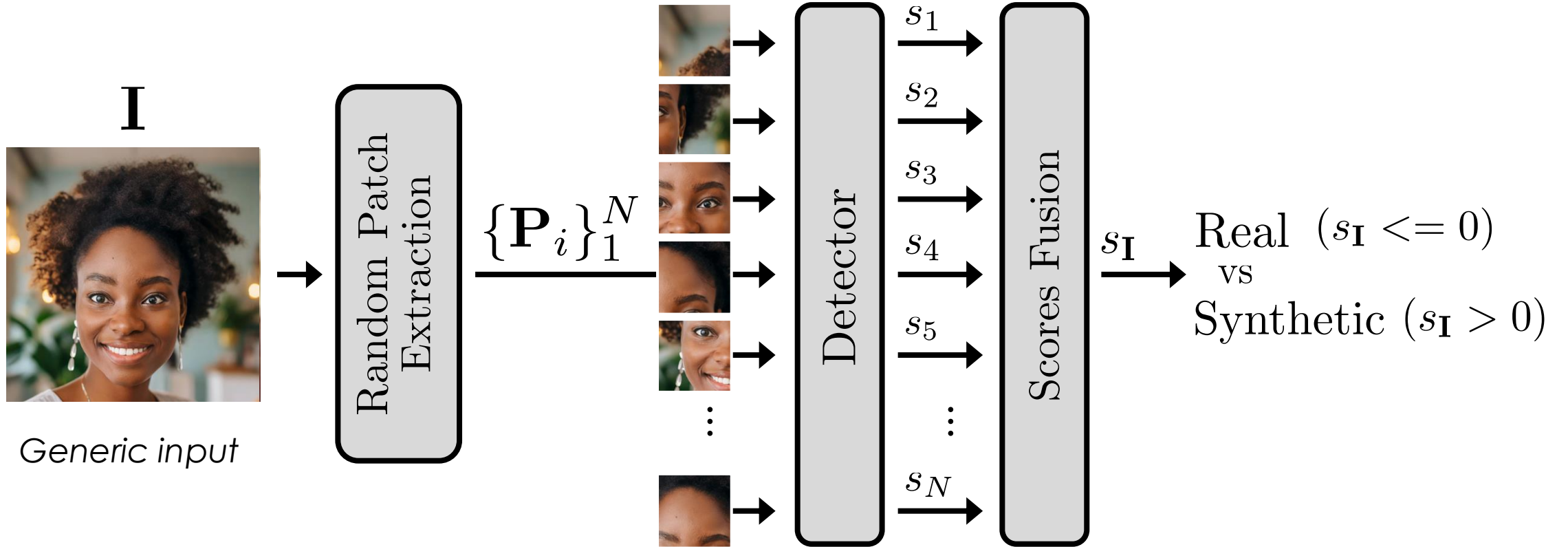
Deepfake *images*:

Specific traces left during the image generation process

- **Generator** *network architecture*
 - *Upsampling operations*
 - *Convolution operations*
 - *Latent space structure*
- **Biases** *inherited from the training data*

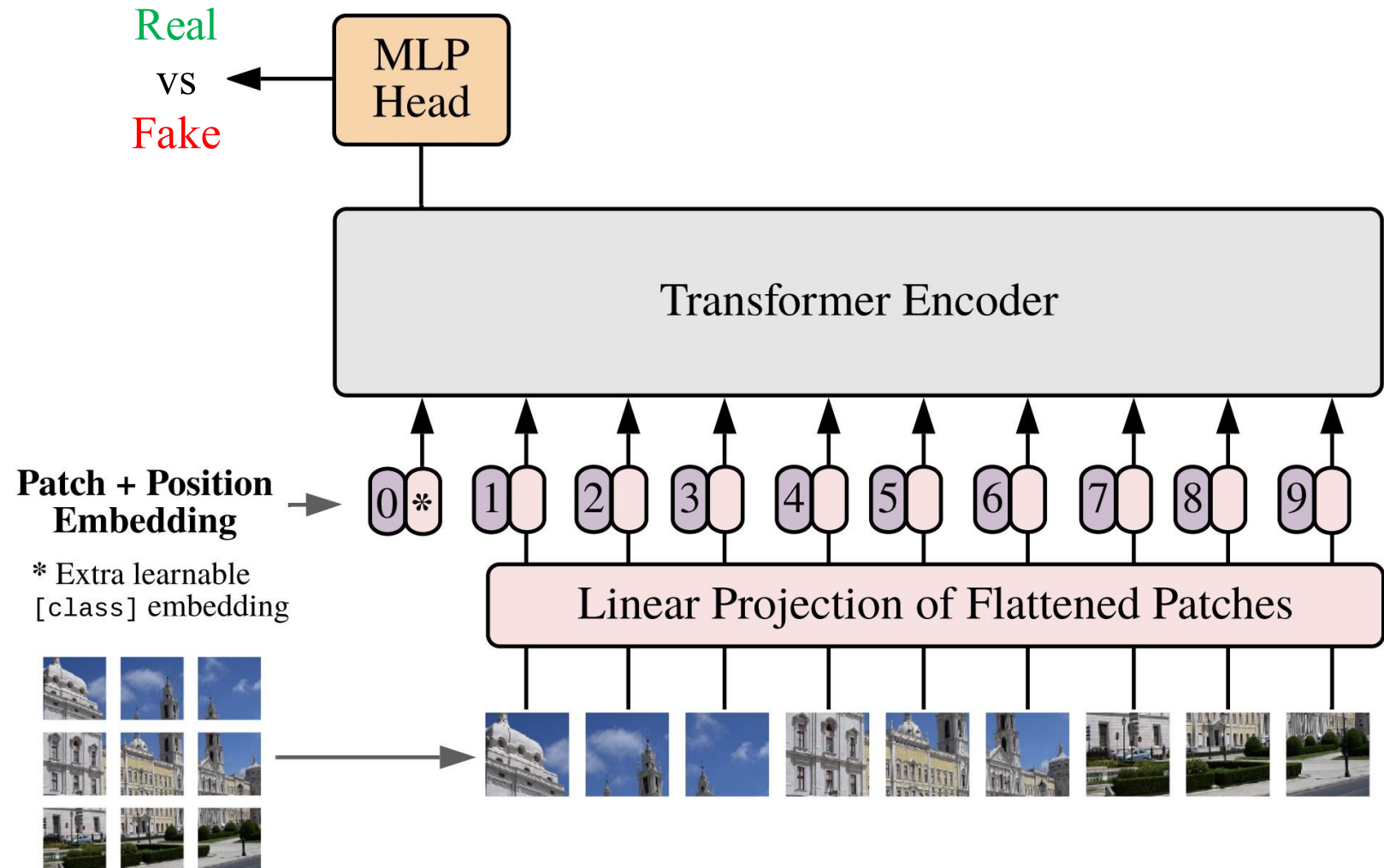
Low level traces: patch-based detection (CNNs or ViTs)

CNN-based *detector*



Low level traces: patch-based detection (CNNs or ViTs)

ViT-based detector



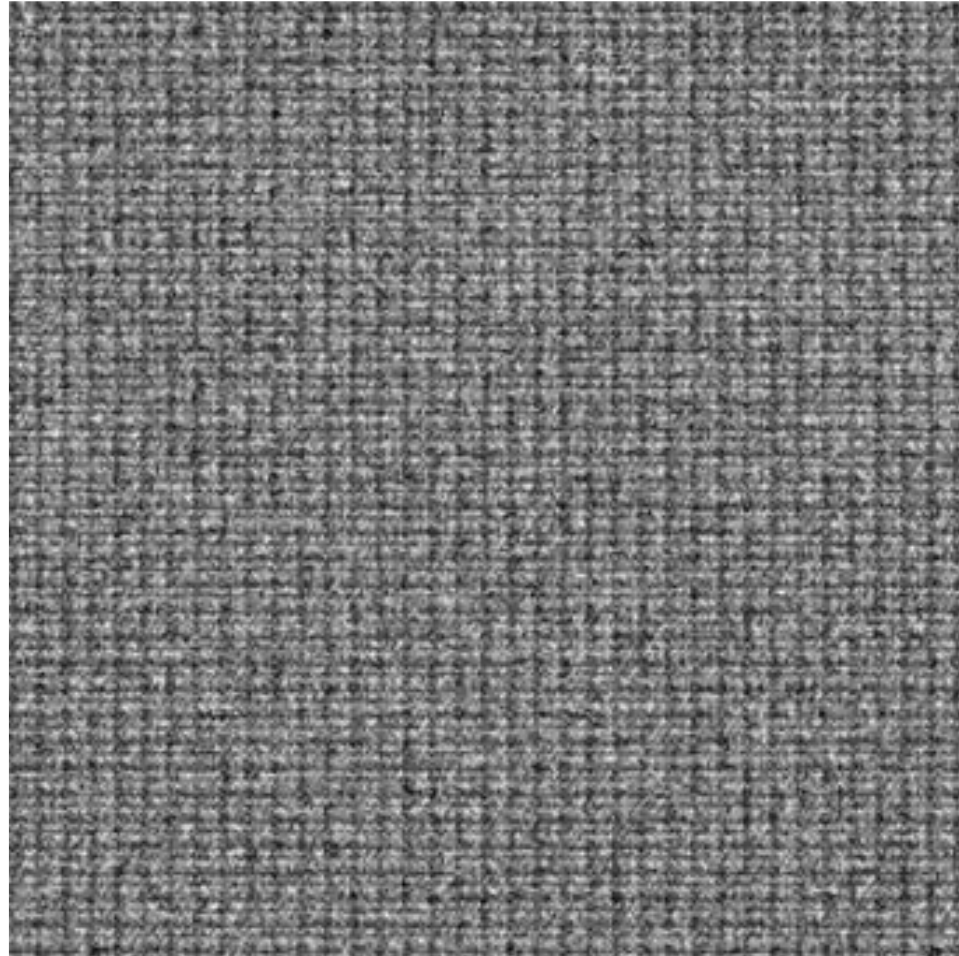
Low level traces: noise residual extraction

Generic input

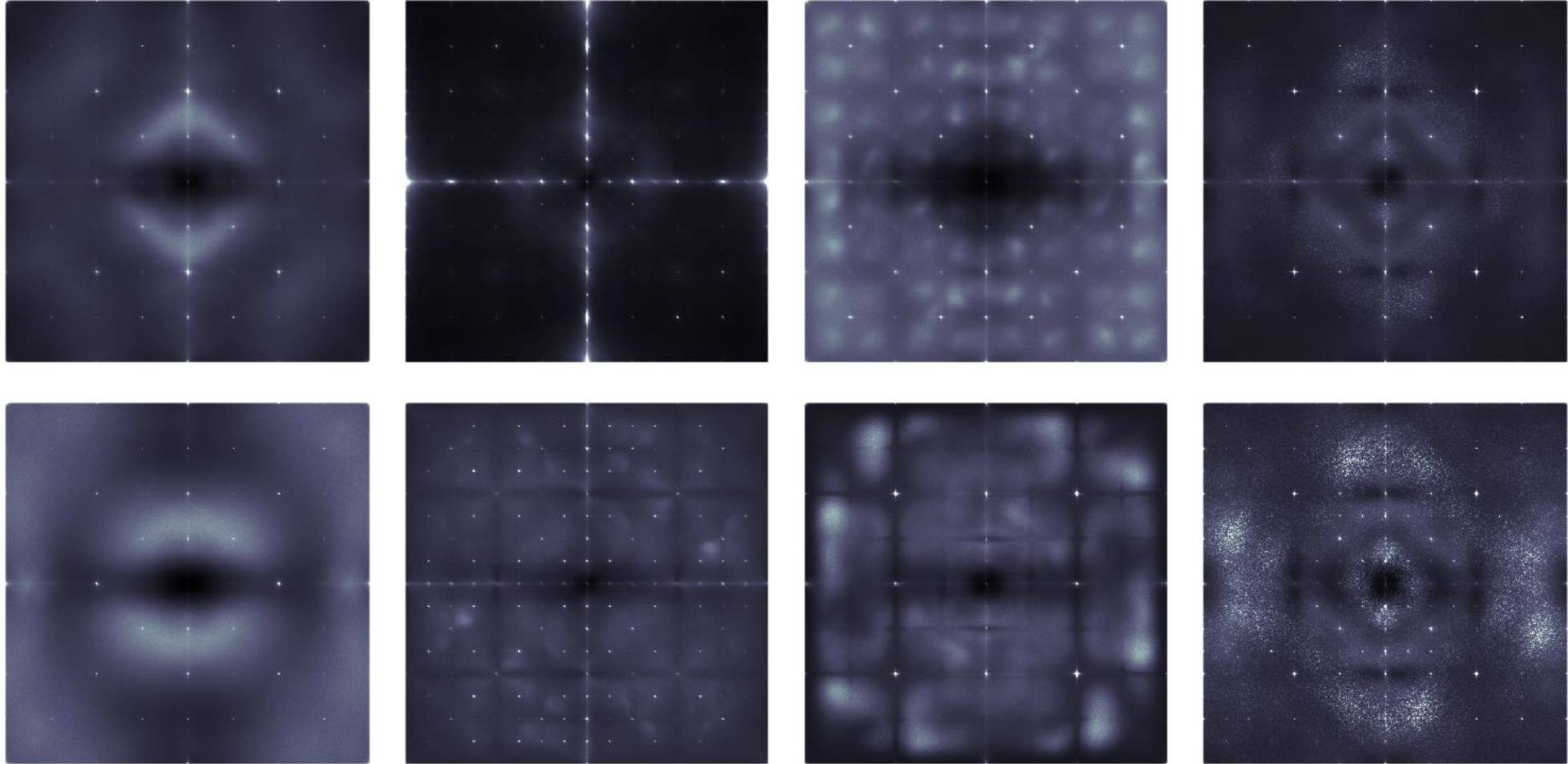


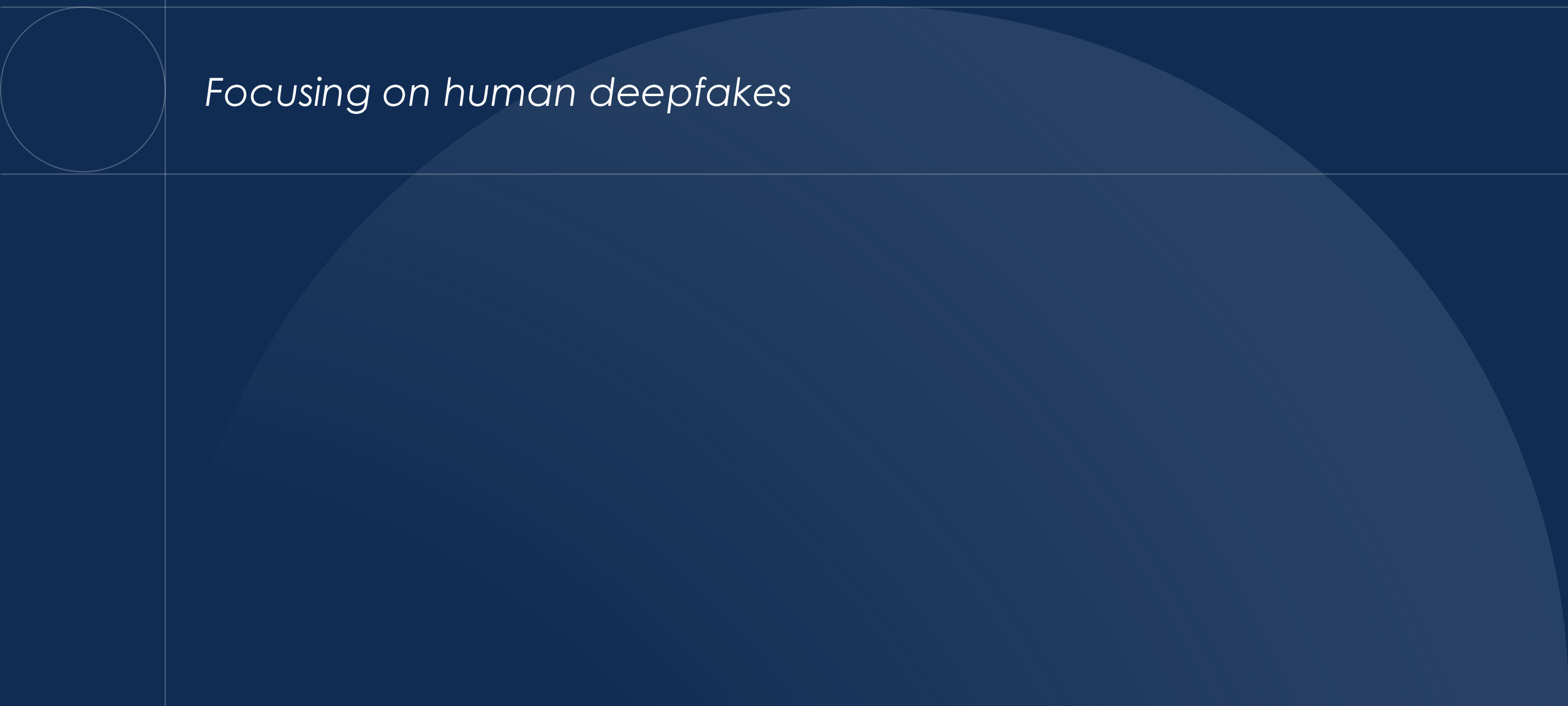
Low level traces: noise residual extraction

Noise residual



Low level traces: noise residual extraction – Fourier spectrum





Focusing on human deepfakes

Human deepfakes



Speech deepfakes: a growing threat

 *Voice Fraud & Social Engineering*

 *Political Deepfakes*

 *Breaking Voice Biometrics*

 *Erosion of Trust*

Finance worker pays out \$25 million after video call with deepfake 'chief financial officer'



By Heather Chen and Kathleen Magramo, CNN

🕒 2 min read · Published 2:31 AM EST, Sun February 4, 2024

Fake Biden robocall tells voters to skip New Hampshire primary election

23 January 2024

Max Matza

BBC News

How 'Voice Cloning' Will Disrupt Customer Verification

Fraudsters can now clone a customer's – or colleague's – voice based on as little as three seconds of audio, according to Sterling Bank. Voice generation is just one of several

Deepfakes and 'the liar's dividend' — are we ready for the future of AI journalism?

By Lucy Sweeney

AI

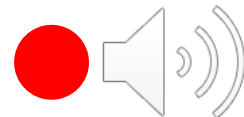
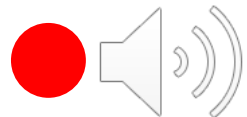
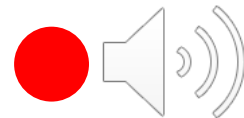
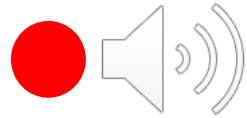
Fri 16 Feb 2024

ve 

es now.

Speech deepfakes: a growing threat

Which is the real human voice?



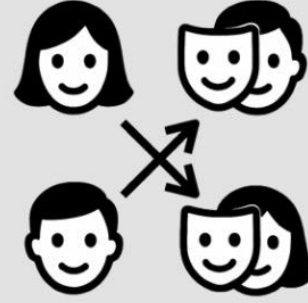
«And on the right, the ripping chisel with which the murders had been committed were exposed to view»



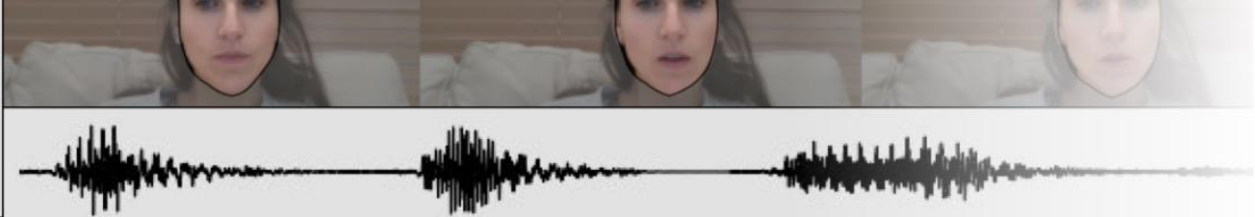
Focusing on human deepfakes

Manipulation techniques

Human deepfakes: face-swap



Face-swap

	Original
	
	<u>Fake face</u>
	Original audio
	
	

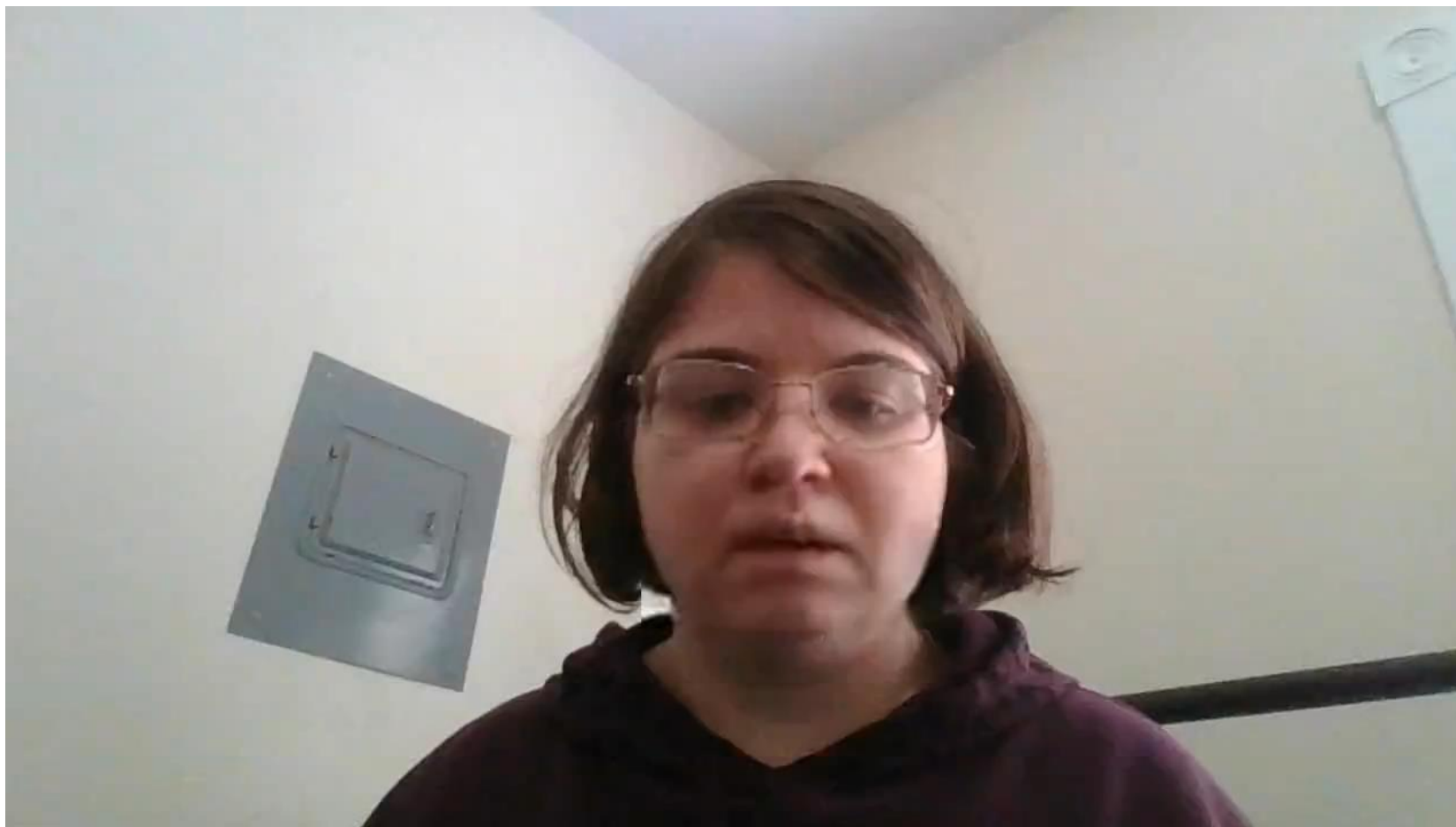
Human deepfakes: face-swap



Human deepfakes: lip-sync

 Lip-sync		Original
		
		<u>Fake</u> mouth <u>Real</u> or <u>Fake</u> audio
		

Human deepfakes: lip-sync



Human deepfakes: avatar



Avatar

Original
(single frame)
Real or Fake face

Animated frame
Real or Fake audio

Human deepfakes: avatar

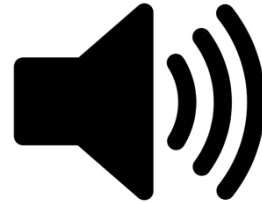


Human deepfake detection



Frame-based approaches

visual modality



Speech-based approaches

audio modality



Multimodal approaches

audio & visual modalities

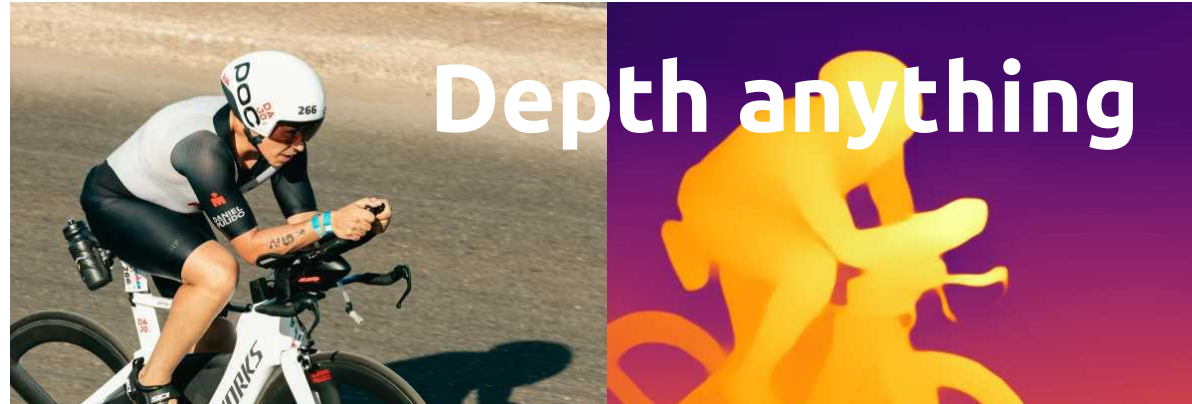
Human deepfake detection

Visual modality: frame-based approaches

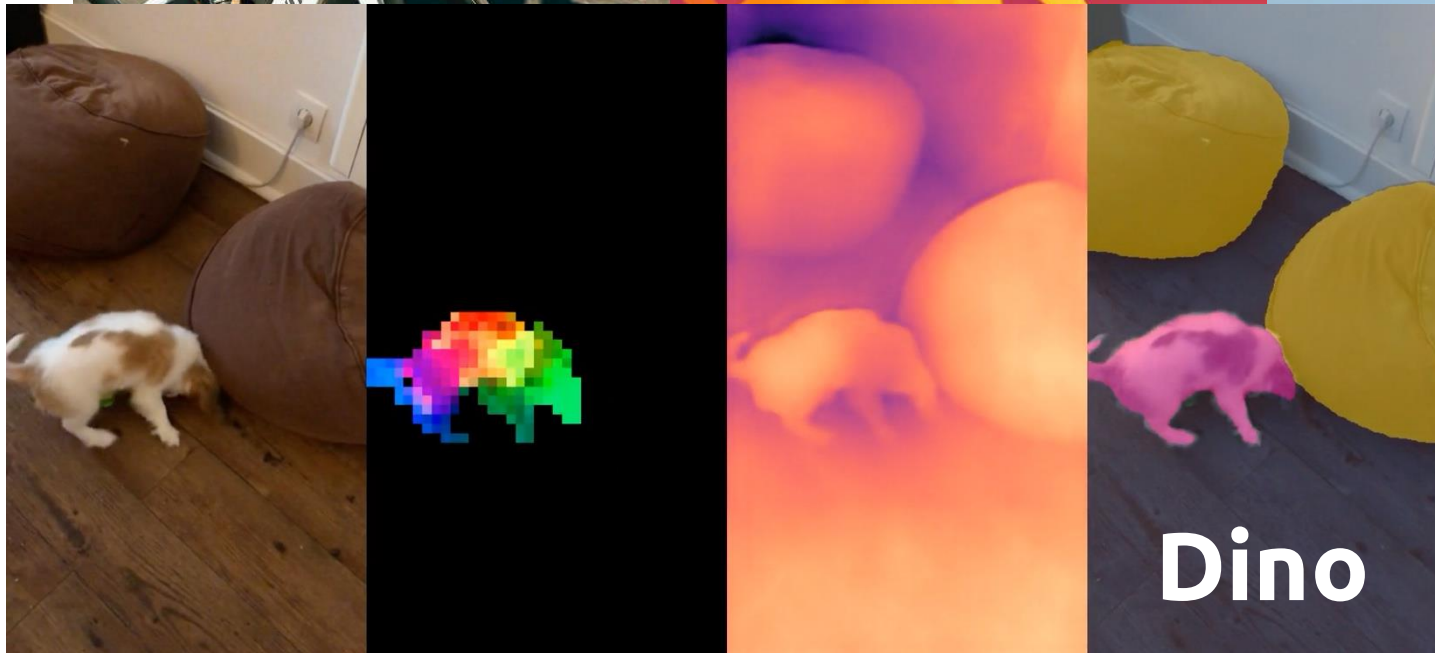




A new trend: detectors based on foundation models



GOAL: extract powerful representations
for many downstream tasks





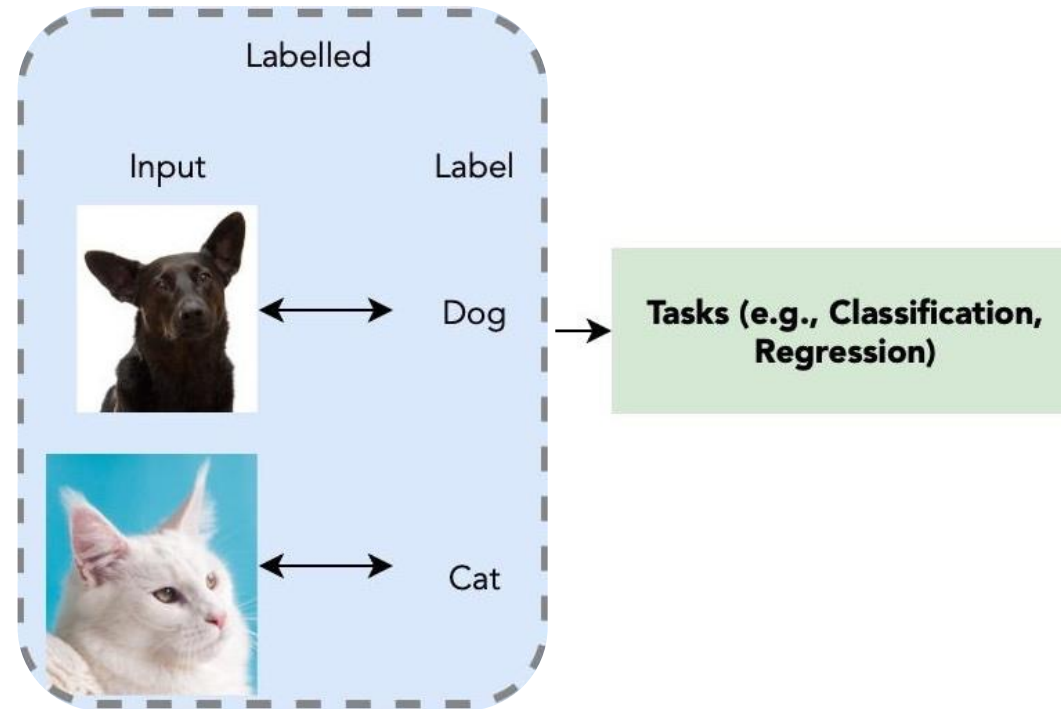
A new trend: detectors based on foundation models

*Foundation models are usually trained in **self supervised** fashion*



A new trend: detectors based on foundation models

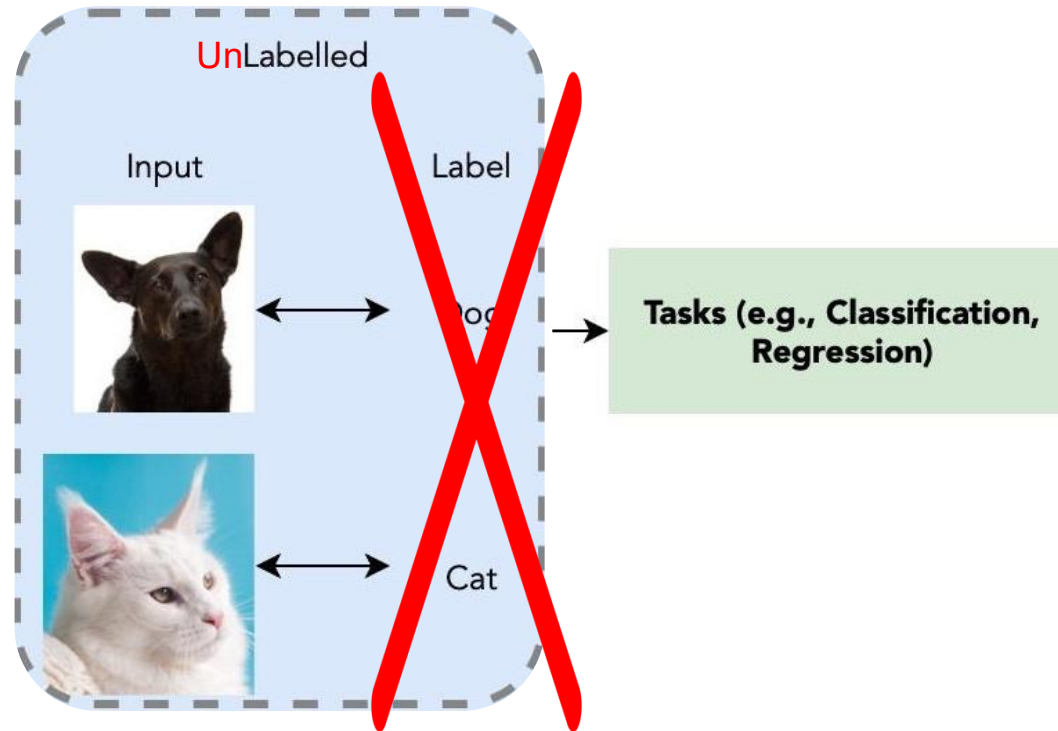
*Foundation models are usually trained in **self supervised** fashion*





A new trend: detectors based on foundation models

*Foundation models are usually trained in **self supervised** fashion*



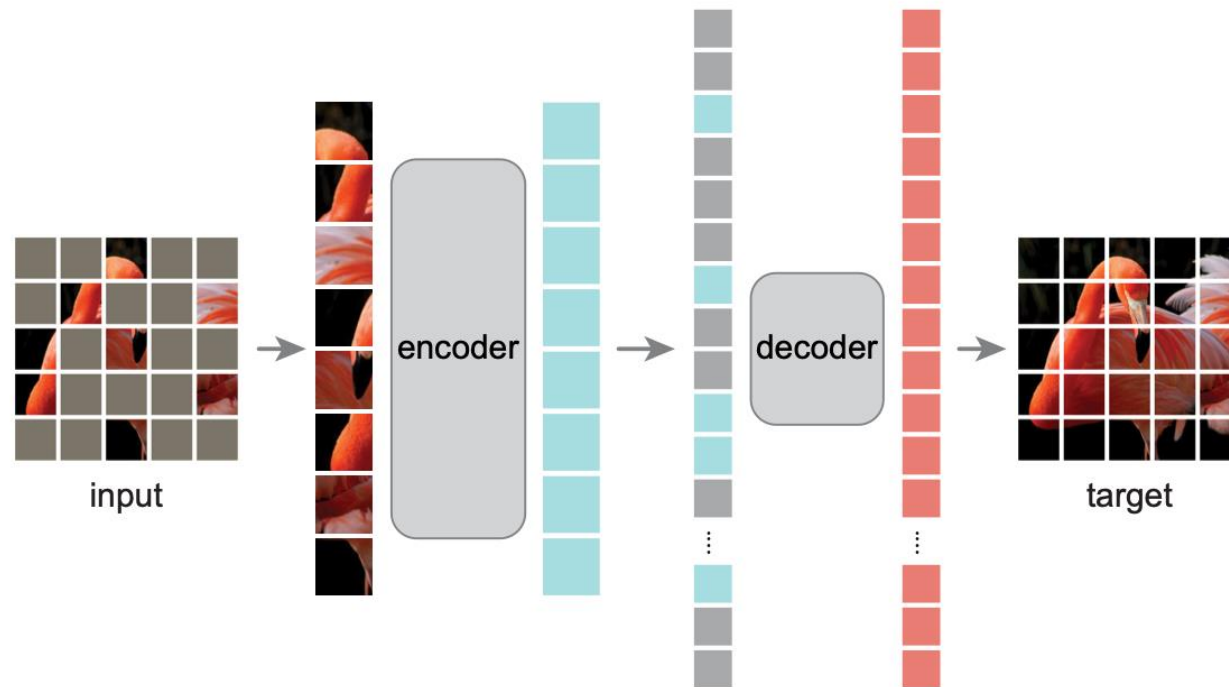


A new trend: detectors based on foundation models

*Foundation models are usually trained in **self supervised** fashion*

Self supervised learning:

- **Masked autoencoder**



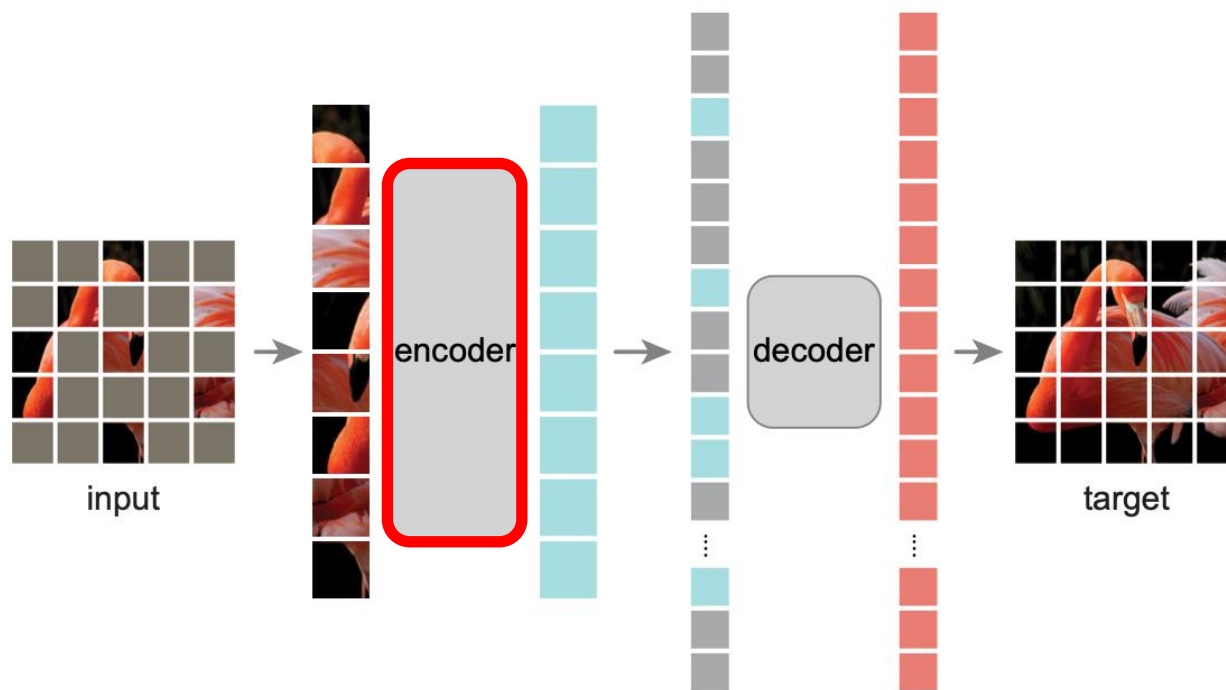


A new trend: detectors based on foundation models

*Foundation models are usually trained in **self supervised** fashion*

Self supervised learning:

- **Masked autoencoder**

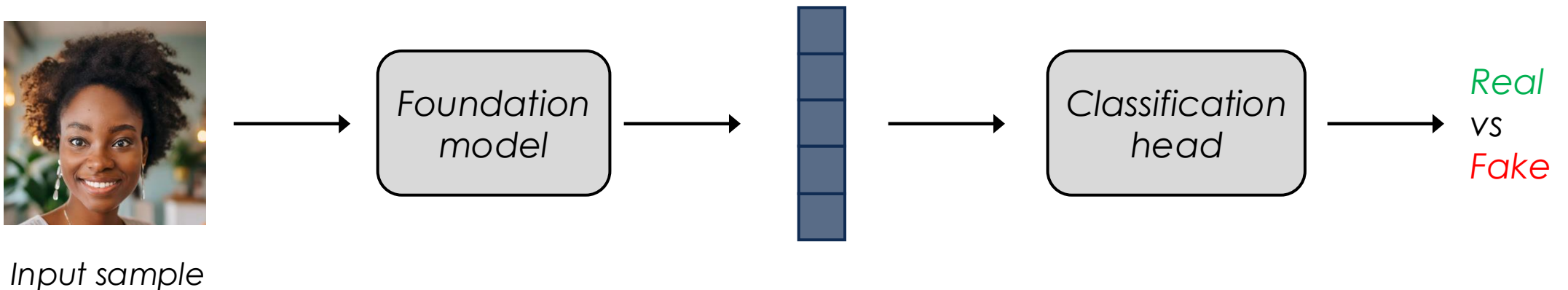




A new trend: detectors based on foundation models

The first block of detectors consist of a **vision foundation model**

- the **foundation model** is used to **extract powerful data representations**
- the model is **fine-tuned** for the deepfake detection task





A new trend: detectors based on foundation models



Dino-v2 based approach



Dino-v2 based approach



Dino-v2 based approach



A new trend: detectors based on foundation models



Dino-v2 based approach + **frequency-domain augmentations**

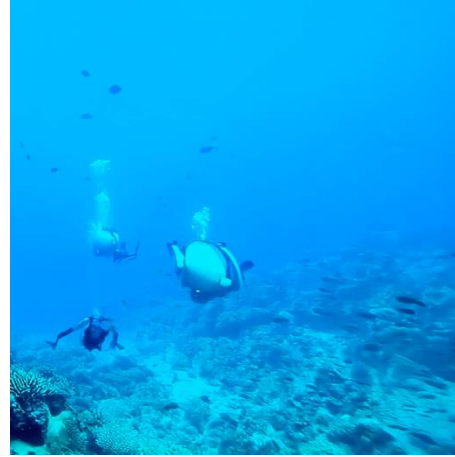


Dino-v2 based approach



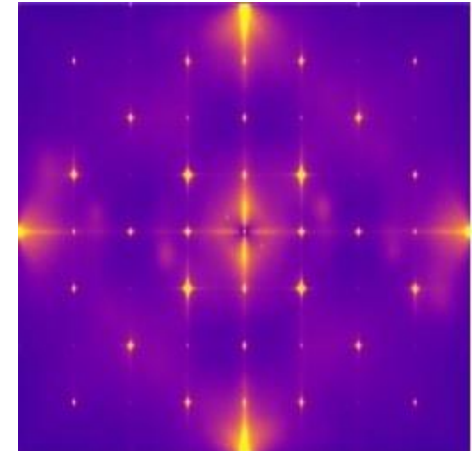
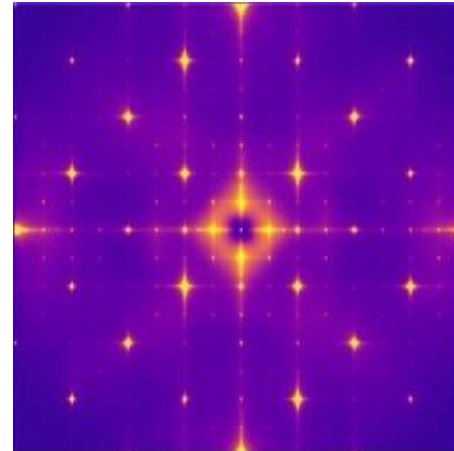
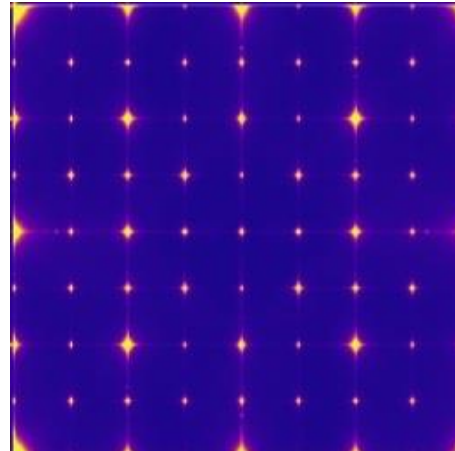
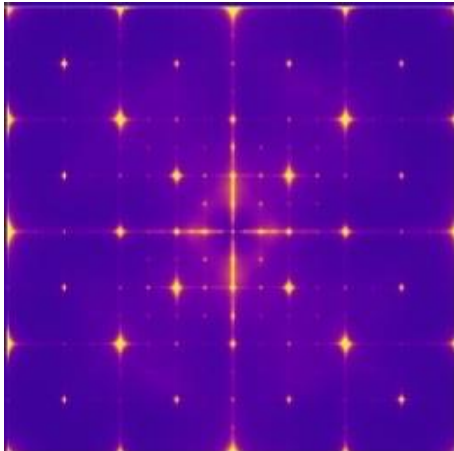
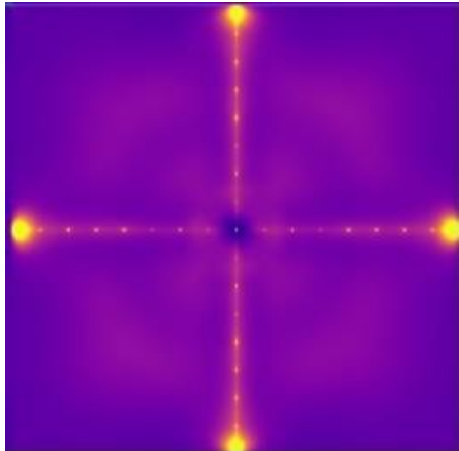
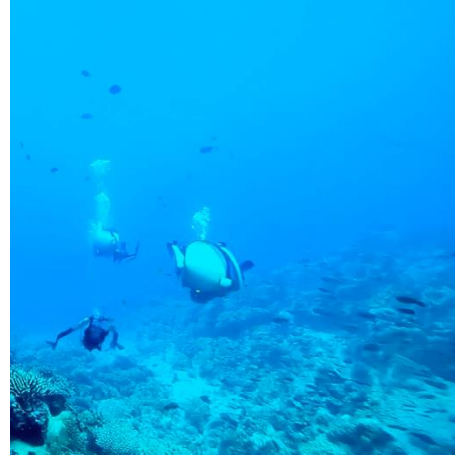
Dino-v2 based approach

Exploiting frequency-based augmentations





Exploiting frequency-based augmentations



Human deepfake detection

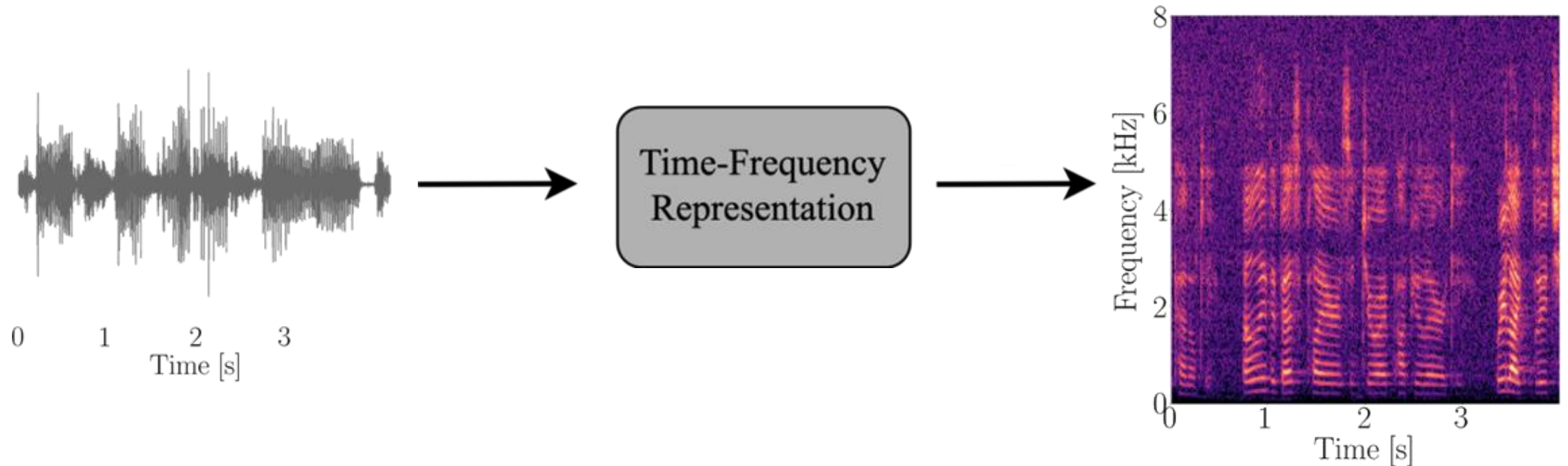
Audio modality: speech-based approaches





Evergreen methods: detectors based on STFT

We can extract **time-frequency representations** from the raw audio and analyze the **spectrogram** as an image

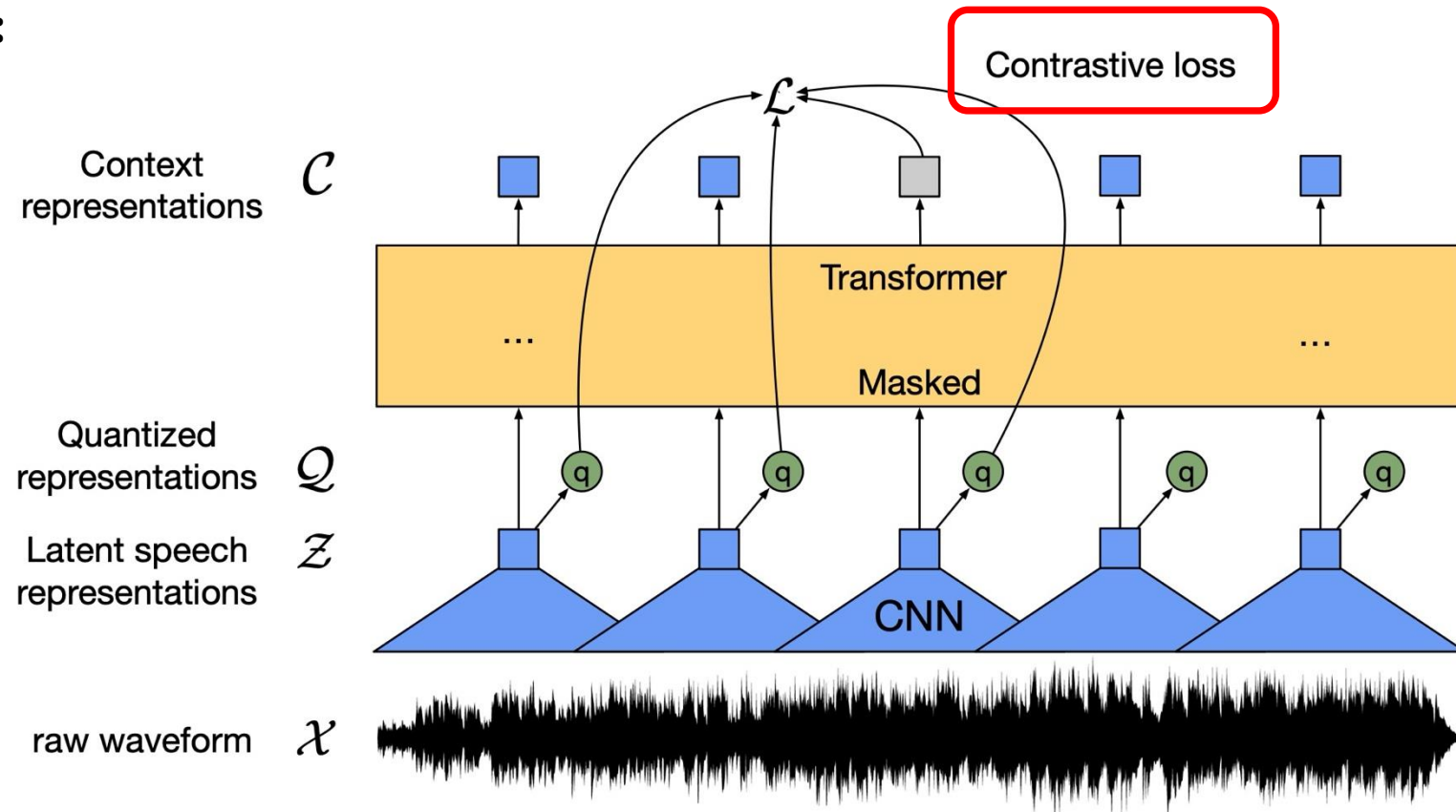




New trends: detectors based on foundation models

Self supervised learning:

- Wav2vec 2.0



Human deepfake detection

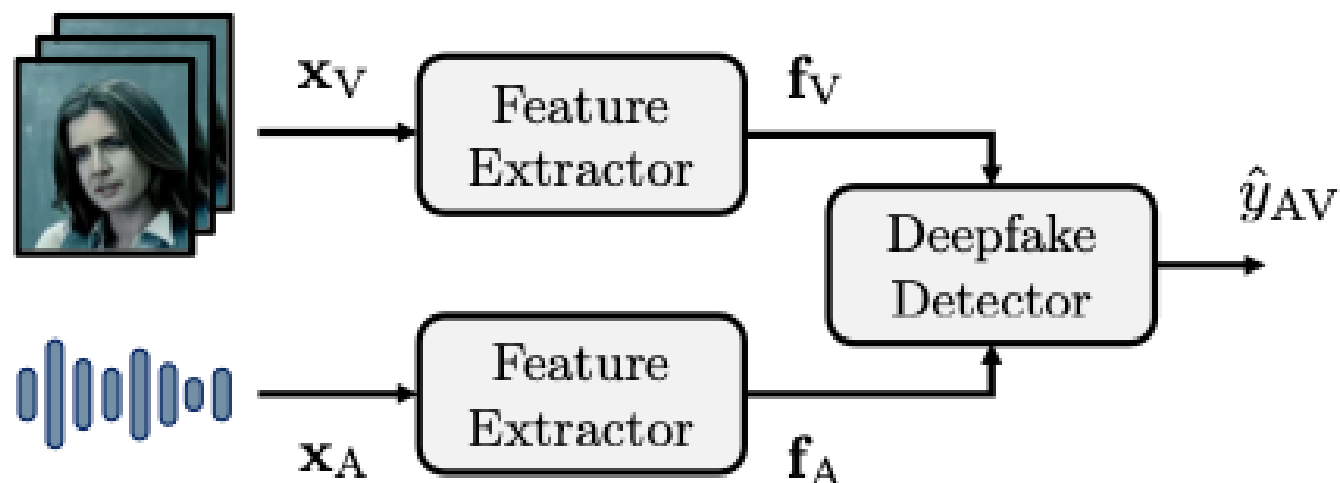
Audio and visual modalities: multimodal approaches





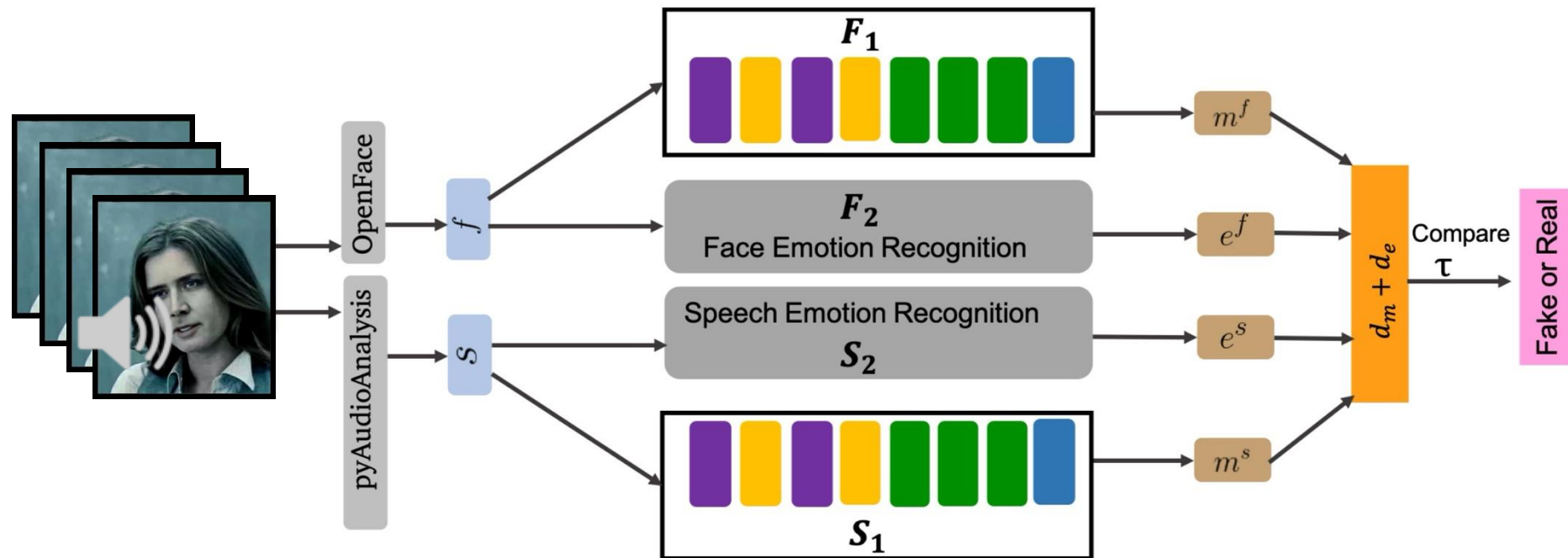
Fusing audio and visual modalities

The typical pipeline includes **two feature extraction branches** and a classification head for the **fusion of the two modalities**





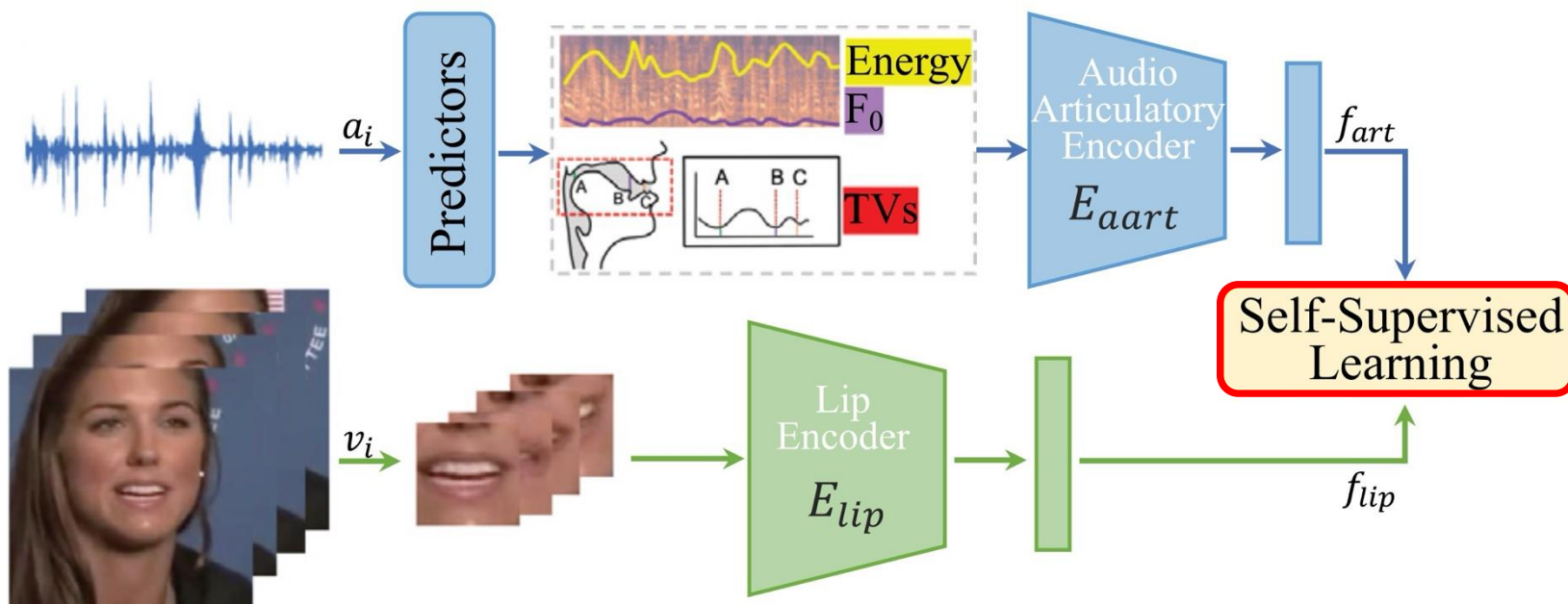
Emotions-based detectors





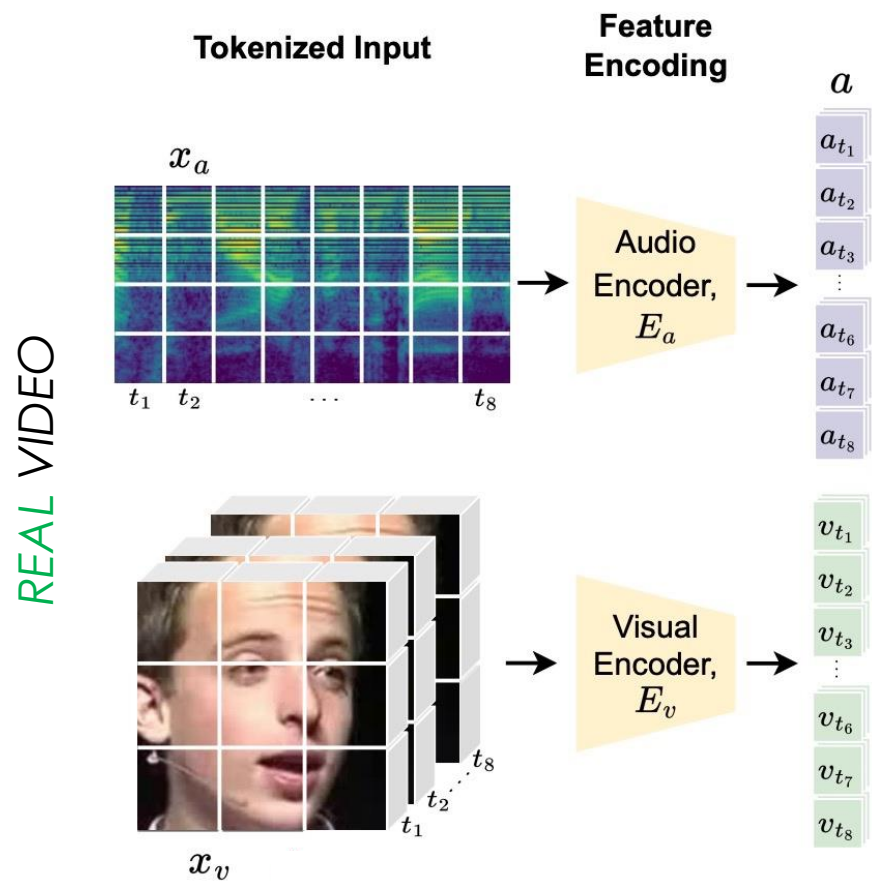
Articulatory representation learning

Self supervised learning: *a confirmed trend*



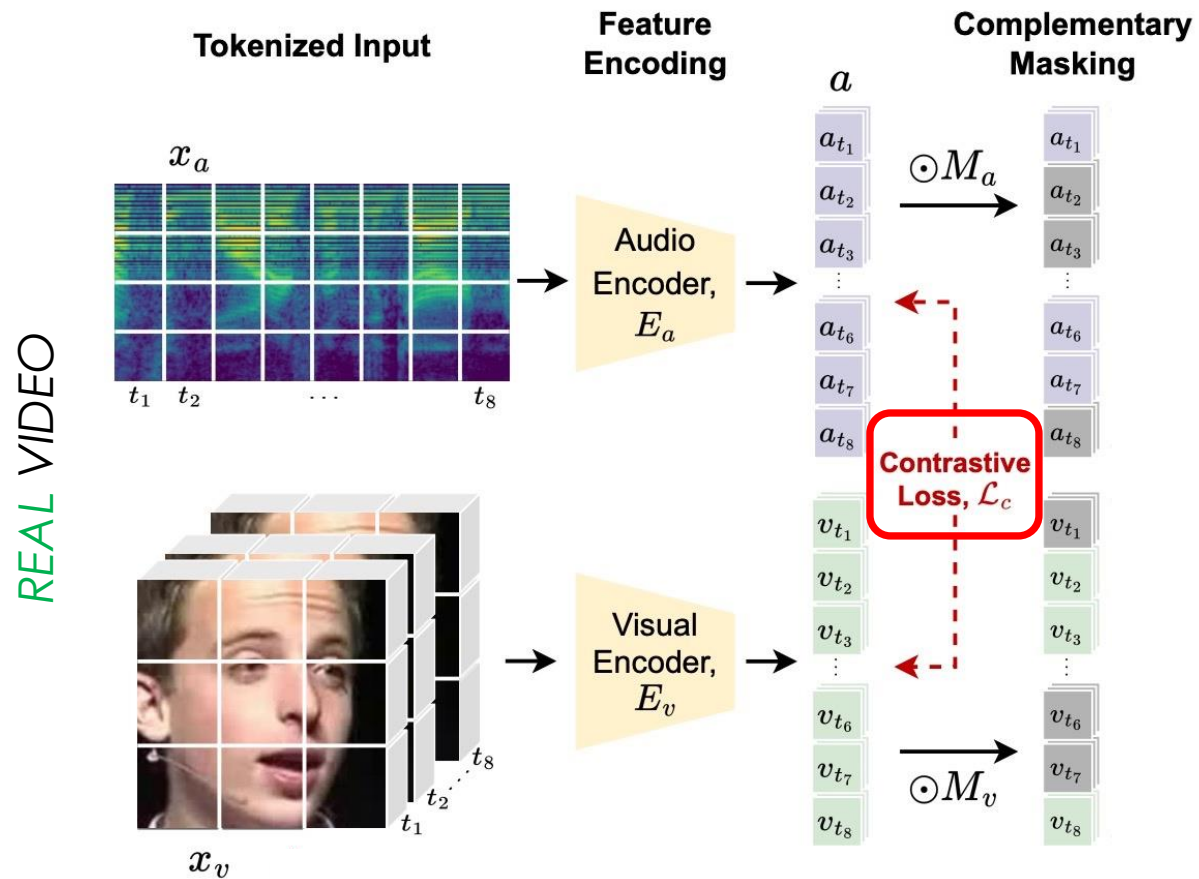


AVFF: Audio-Visual Feature Fusion



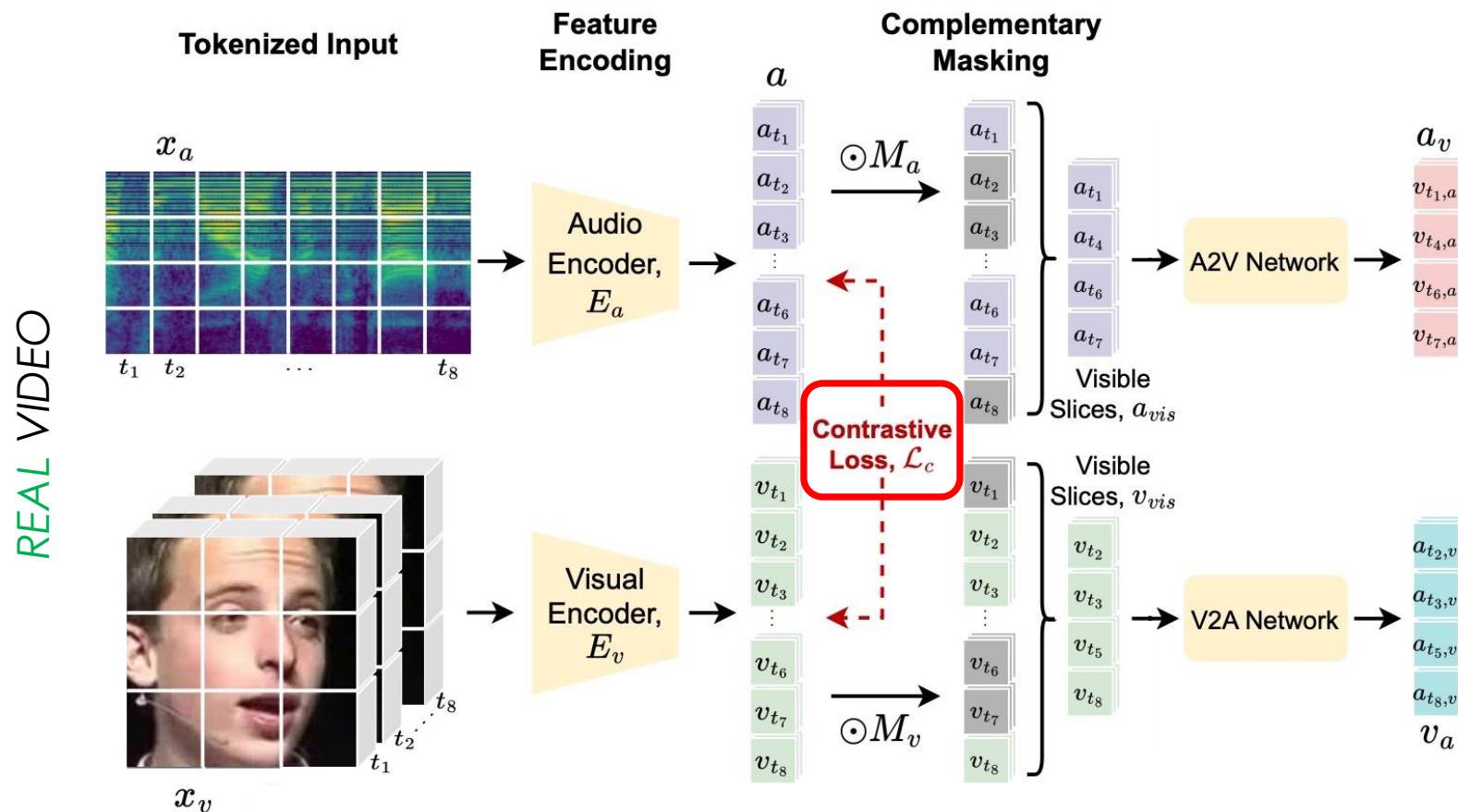


AVFF: Audio-Visual Feature Fusion



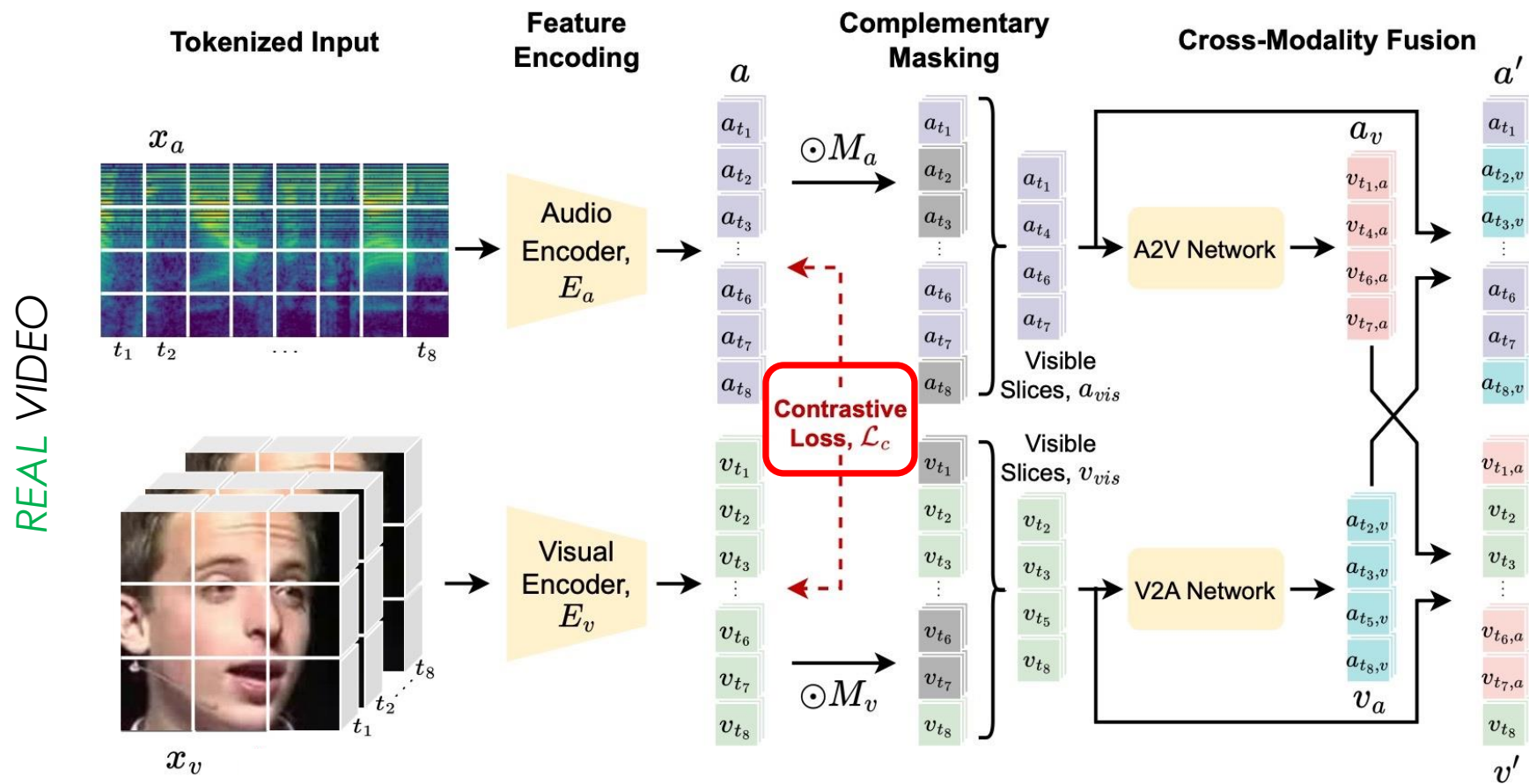


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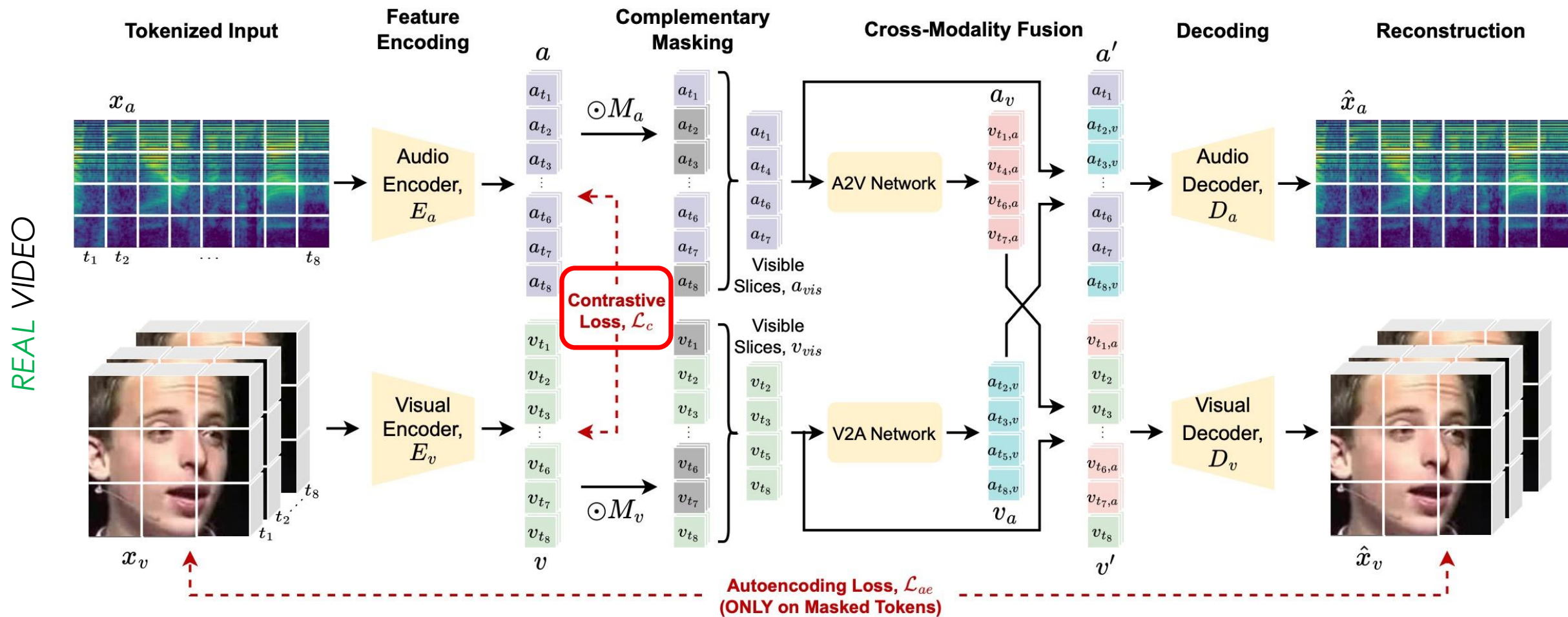


AVFF: Audio-Visual Feature Fusion





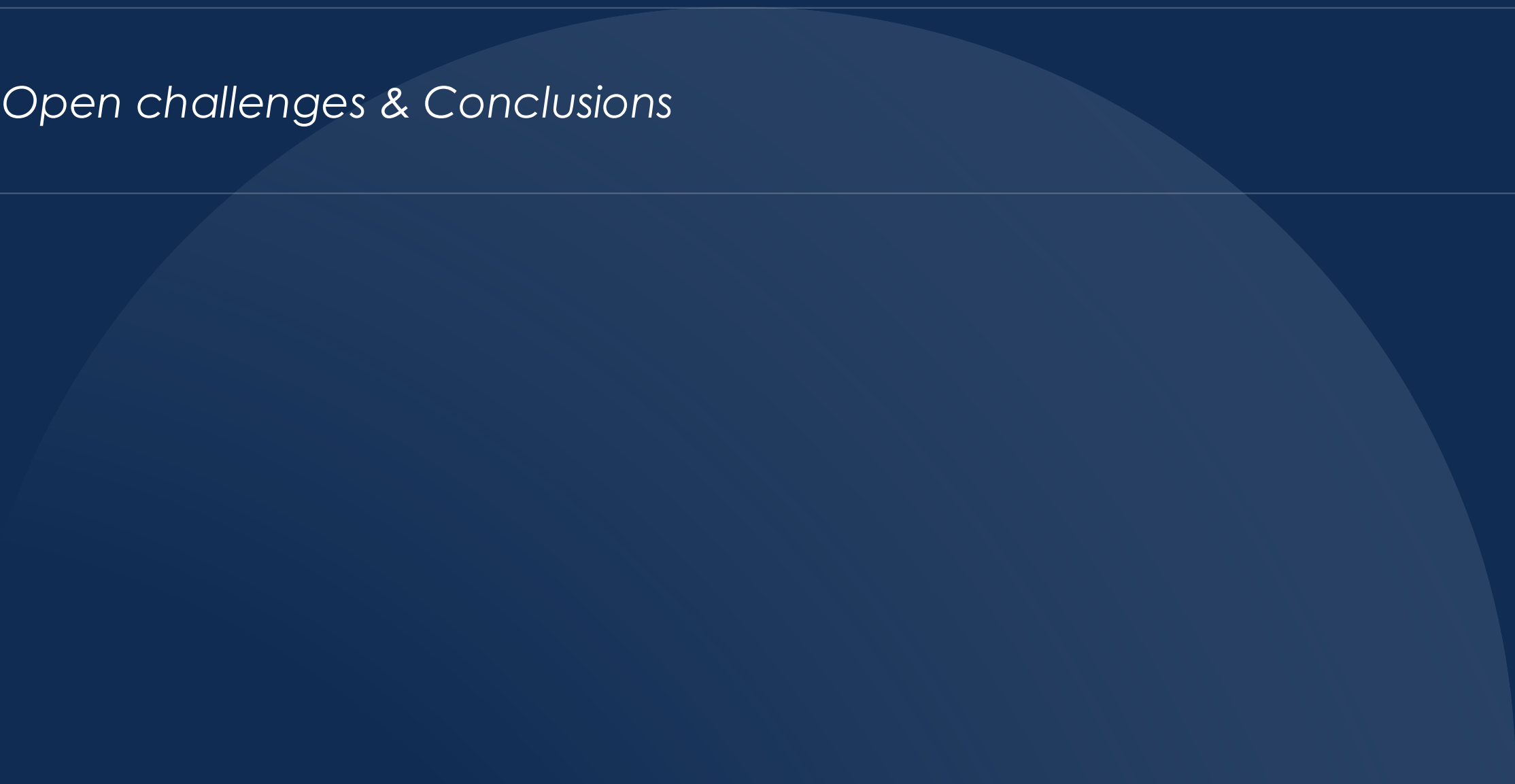
AVFF: Audio-Visual Feature Fusion





Let's see some examples

Notebook time

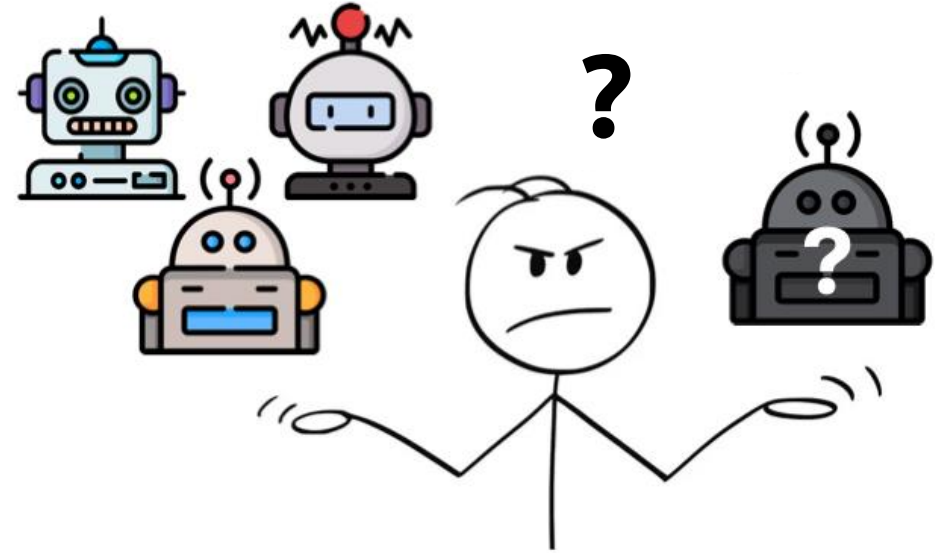


Open challenges & Conclusions

Open challenges

The Generalization Problem

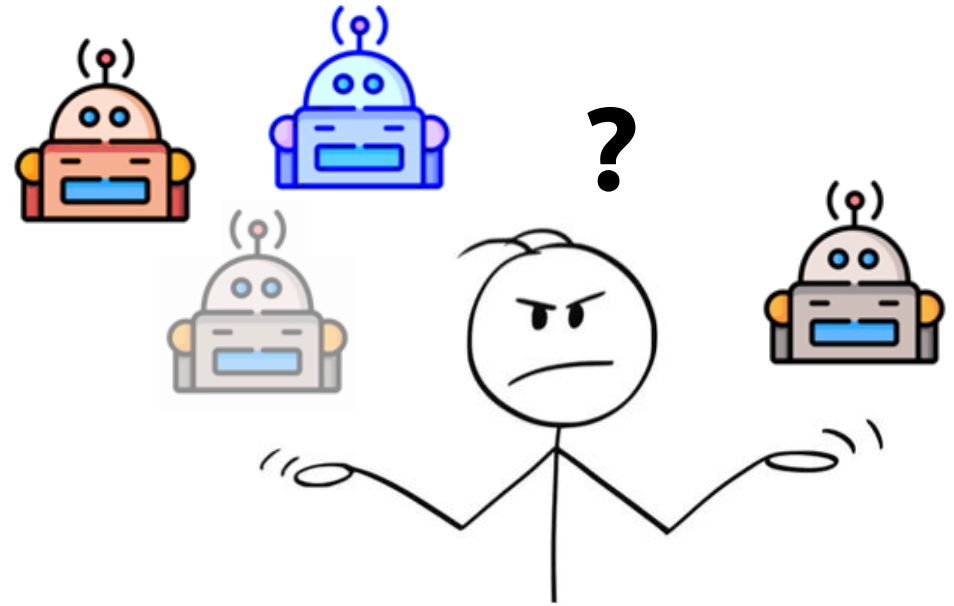
- Existing deepfake detectors lack generalization capabilities on **unknown** data
 - a. Unseen synthetic speech generators
 - b. Out of domain real data
- With the increasing frequency of **new generative models** releases, it is essential for detectors to adapt efficiently



Open challenges

The Robustness Problem

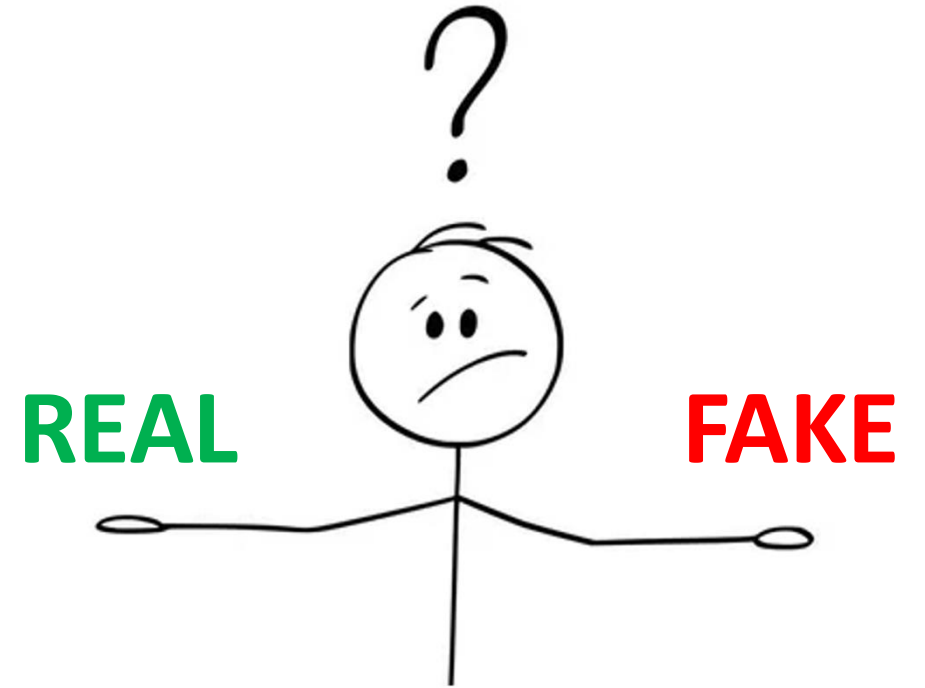
- Existing deepfake detectors lack robustness in **mismatched** conditions
 - a. Post-processed data
 - b. Degraded data
- With the increasing **spread** of synthetic content **online**, it is essential for detectors to adapt efficiently



Open challenges

The Interpretability/Explainability Problem

- Existing deepfake detectors **lack interpretability** in their decision making
 - Detectors are usually employed as **black-boxes**
- The use of detectors' outputs as evidence in **courts of law** is strongly hindered by this lack of interpretability



Conclusions: much work remains to be done



Hey
Gen



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Thanks for the attention

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